The Whitening Advantage: Exact Calibration of Concept-Drift Detectors on Heteroscedastic Streams

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Abstract

Sequential change-point detectors (CUSUM, ADWIN, Page–Hinkley) are calibrated under an i.i.d. hypothesis that financial streams violate through volatility clustering. We quantify the damage and remove it. On the raw stream, the GARCH penalty Γ, the normalized spectral density at zero frequency of the monitored statistic, drives IID-calibrated false-positive rates above 80%. The cure is detector-specific ($\lambda\times\Gamma$ for cumulative statistics, $\epsilon_{\text{cut}}\times\sqrt{\Gamma}$ for window means) and defers detection by the very factor it applies: a *Lethargy Tax*, with a $\sqrt{w\Gamma}$ scaling law for gradual drifts established in a deterministic-drift approximation. On the sign-prediction stream, we prove a whitening property: under conditionally symmetric innovations, the binary error stream of any non-anticipative classifier is i.i.d. Bernoulli$(1/2)$ whatever the volatility dynamics, so calibration becomes exact, distribution-free, and detector-agnostic. What no monitor escapes is the information budget: a *Sharpe ceiling* $\text{ADD}_{\min} \geq 504\ln(1/\alpha_0)/\text{SR}^2$, at admissible false-alarm level $\alpha_0 = 1/\text{ARL}_0$, bounds detection for any monitor of the *unconditional* daily-return stream, and on four ETFs (2000–2025) it puts 80% of dated directional regimes out of budget even at the loosest calibration. A look-ahead oracle backtest locates the boundary: granted look-ahead GARCH parameters, a variance-aware monitor detects the 2020 COVID-19 crash in 3–16 trading days, while the crash exhausts the sign stream's own budget—the price of binarization, paid exactly where location and variance shocks coincide. Speed, finally, is settled by Pitman efficiency, not by volatility: at matched false-alarm rate the sign filter trails correct parametric standardization by $1/(4f_z(0)^2)$ ($\pi/2$ under Gaussianity) and overtakes it below a measured $\nu^\star \approx 4.9$ degrees of freedom, precisely where parametric estimation is most fragile.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → **Time series analysis**; • **Information systems** → *Data streams*.

Keywords

concept drift, GARCH, volatility clustering, whitening filter, sequential detection, financial time series, scaling law

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1 Introduction

Financial machine-learning systems run on streams marked by volatility clustering, heavy tails, and near-unit-root persistence [13, 37]—stylized facts that break the i.i.d. assumption underlying standard concept-drift detectors [19, 23]. The streaming literature still evaluates detectors almost exclusively on i.i.d. synthetic generators; the few financial studies ran DDM, EDDM, and ADWIN on S&P 500 data without modeling the volatility dynamics [9, 26], while econometric sequential monitoring presupposes a fitted GARCH model [21, 34]. How drift detectors behave between these two worlds—heteroscedastic streams, no parametric crutch—has remained open.

This paper closes that gap with four formal contributions, controlled validation on stationary GARCH(1,1) regimes calibrated via the ProteuS generator [35], and an in-the-wild backtest on daily ETF streams (FirstRate Data) spanning the 2020 COVID-19 variance shock. Our central object—the exact fair-coin null of a sign-error stream under conditional symmetry—elevates a classical robustness of sign statistics [7, 12] into a streaming calibration recipe. What is new is not that signs are robust, but that the resulting null is an *exact, finite-sample, distribution-free* pivot with no specification step, and therefore supports anytime-valid monitoring that no fitted-volatility route can offer.

Architectural premise and scope. The classifier itself is a generic proxy: the whitening property holds for *any* non-anticipative classifier, and the invariance comes entirely from the conditional binarization of the error stream, never from the learning dynamics. We instantiate it with a deliberately *non-adaptive* Hoeffding Tree, which keeps the drift signal available to an external monitor. An adaptive learner would partially absorb the change and share the drift signal with the detector; quantifying that split—how much power the monitor cedes, and how the learner's recovery speed trades against detection delay—is a distinct problem that a single ablation would not settle, and we do not address it here. The guarantee itself is unaffected: an adaptive learner remains $\mathcal{F}_{t-1}$-measurable, so Proposition 3.1 continues to hold; what changes is the alternative, not the null. All results below are therefore stated for the frozen architecture.

Contributions.

(C1) The GARCH penalty factor Γ and its detector-dependent cure. We define the variance-inflation factor $\Gamma = 1 +$

$2\sum_{k\geq 1}\rho_e(k)$ of the monitored statistic via the FCLT for β-mixing processes; uncalibrated CUSUM and ADWIN false-positive rates approach or exceed 80% once $\Gamma \gtrsim 20$, and the corrective inflation is detector-dependent: $\lambda \times \Gamma$ for cumulative statistics (CUSUM, PHT; Siegmund regime), $\epsilon_{\text{cut}} \times \sqrt{\Gamma}$ for the window-mean ADWIN.

(C2) Two-regime scaling law for gradual scale drifts. In a deterministic-drift approximation, the DATA delay under a linear-ramp variance drift of width w, at the recalibrated threshold $\lambda_{\text{iid}}\Gamma$, scales as $K\sqrt{w\Gamma} + \rho w$ above an explicit crossover and saturates below it, where $\rho = \delta_P/\Delta\mu_{\max}$ is the detector's dead band. The additive ρw—the fraction of the ramp the tolerance blinds the monitor to—is what lifts the measured exponent above 1/2, and it eventually dominates: on an arbitrarily slow ramp the delay is set by the tolerance alone, independently of Γ and of the threshold. Five penalties at a common horizon match the law to a 5% median error with no fitted constant.

(C3) The Sign-Task Whitening Property. We prove that on a sign-prediction task with conditionally symmetric innovations, the binary error stream of *any* non-anticipative classifier is i.i.d. Bernoulli(1/2)—structurally insensitive to Γ—validate it via Ljung–Box tests on 360 independent streams and flat delays across $\Gamma \in [1, 200]$, and map its boundaries: exact past the loss of the fourth moment, broken by non-median thresholds or continuous losses, free of the specification risk of econometric standardization.

(C4) Location/scale orthogonality and the Sharpe ceiling. We prove the sign sensor isolates the location component at first order and is exactly blind to scale (Proposition 3.3), with closed-form relative efficiency $1/(4f_z(0)^2)$ against correct parametric standardization (Proposition 3.5); we derive the delay floor for any monitor of the *unconditional* daily-return stream, $\text{ADD}_{\min} \geq 504\ln(1/\alpha_0)/\text{SR}^2$ trading days at false-alarm level α_0 (Corollary 3.6), and show on a 25-year, four-ETF census that it exceeds the phase duration for 80% of dated directional episodes, a look-ahead oracle backtest (Section 3.7) delimiting the bound's scope.

2 Background and the GARCH Penalty Factor

2.1 GARCH(1,1): Clustering, Mixing, and the FCLT

The GARCH(1,1) model [6, 16] captures volatility clustering via:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{1}$$

where (z_t) are i.i.d. innovations with zero mean, unit variance, and a continuous distribution symmetric about zero (Gaussian, or the standardized Student-t_ν of our experiments, t_7 unless stated otherwise), and $\alpha + \beta < 1$ ensures second-order stationarity. Provided $\mathbb{E}[\varepsilon_t^4] < \infty$ (equivalently $\mathbb{E}[(\alpha z_t^2 + \beta)^2] < 1$), the squared residuals ε_t^2 follow an ARMA(1,1) with AR parameter $\alpha + \beta$ [17], hence geometric ACF decay with the persistence $\alpha + \beta$ as the signature of volatility clustering [15].

A defining property of the model is that (ε_t) is a *martingale difference sequence*, $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$, with $\mathcal{F}_{t-1}$ the natural filtration: returns are serially uncorrelated, and volatility clustering lives only in the even moments. Under stationarity (ε_t), hence (ε_t^2), is geometrically β-mixing [8]; by the FCLT for strongly mixing sequences [20, 25], provided $\mathbb{E}|\varepsilon_t|^{4+\delta} < \infty$, partial sums of ε_t^2 carry a variance inflated by the long-run factor Γ of Section 2.2.

Terminology: the four monitored statistics. Four monitoring pipelines recur, each tagged by its statistic and set out in Table 1: DATA (the raw squared residual, a scale sensor), ECO-L1 (the QMLE-standardized residual, the parametric location monitor), RECALIB (its square, used only in the iso-FPR race of Section 3.3), and CONCEPT (the binary sign-error stream of a non-adaptive classifier, the subject of Proposition 3.1). "The DATA pipeline" always means the raw squared sensor, never ECO-L1.

2.2 The GARCH Penalty Factor

Because raw returns are serially uncorrelated (Section 2.1), volatility clustering can only penalize a sequential detector through a *volatility-sensitive* monitored statistic. Detecting structural breaks in variance is a primary goal in financial monitoring [1, 21], making the scale sensor a necessary baseline before isolating location drifts. Throughout the DATA pipeline, the monitored statistic is therefore the standardized squared-residual stream

$$e_t := \frac{\varepsilon_t^2 - \widehat{\mu}_{\varepsilon^2}}{\widehat{\sigma}_{\varepsilon^2}}, \tag{2}$$

where $\widehat{\mu}_{\varepsilon^2}$ and $\widehat{\sigma}_{\varepsilon^2}$ are the mean and standard deviation of ε_t^2 estimated on a drift-free warm-up window. We define the **GARCH penalty factor** Γ as the normalized spectral density at zero frequency of the monitored statistic [27]:

$$\Gamma := 1 + 2\sum_{k=1}^{\infty}\rho_e(k), \tag{3}$$

where $\rho_e(k) = \text{Corr}(e_t, e_{t+k})$. Under a stationary GARCH(1,1) with $\mathbb{E}[(\alpha z_t^2 + \beta)^2] < 1$ (equivalently $\kappa_z\,\alpha^2 + 2\alpha\beta + \beta^2 < 1$, where $\kappa_z := \mathbb{E}[z_t^4]$ is the innovation kurtosis), the ARMA(1,1) representation of the squares (Section 2.1) implies the geometric ACF $\rho_e(k) = \rho_e(1)\,(\alpha+\beta)^{k-1}$, yielding the closed form [6, 17]:

$$\Gamma = 1 + \frac{2\rho_e(1)}{1-\alpha-\beta}, \qquad \rho_e(1) = \frac{\alpha\big(1-\beta(\alpha+\beta)\big)}{1-2\alpha\beta-\beta^2}. \tag{4}$$

Kurtosis-free ACF.. Although $\mathbb{E}[\varepsilon_t^4] < \infty$ is required for Γ to be finite, Eq. (4) is itself *kurtosis-free*: κ_z cancels inside $\rho_e(1)$, so it holds verbatim for our standardized Student-t_7 innovations ($\kappa_z = 5$). Operationally Γ inflates the variance of partial sums, $\text{Var}(\sum_{i\leq n} e_i) \sim n\sigma_e^2\Gamma$: short-term memory surfaces as macroscopic uncertainty and penalizes any monitor calibrated at $\Gamma = 1$. Table 2 lists the regimes tested.

2.3 The False Positive Explosion

Partial sums of the monitored statistic e_t of Eq. (2) carry effective variance $n\Gamma$ instead of n. To show what ignoring this costs, we deploy two IID-calibrated detectors on DATA streams from stationary GARCH processes under H_0 (300 streams of 5000 steps per grid point): a StrictCUSUM[1] ($\lambda_{\text{iid}} = 65$, $\delta_P = 0.5$) and an ADWIN-type window-mean detector [5].

[1]The one-sided CUSUM recursion $S_t = \max(0,\, S_{t-1} + e_t - \delta_P)$ [29], alarming as soon as $S_t > \lambda$, evaluated at every step.

Table 1: The four monitoring pipelines: statistic, null calibration, sensor type (Proposition 3.3), and sections of use. DATA/RECALIB read scale; ECO-L1/CONCEPT read location.

Pipeline	Monitored statistic	Null calibration	Sensor	Used in
DATA	$e_t = (\varepsilon_t^2 - \hat{\mu}_{\varepsilon^2})/\hat{\sigma}_{\varepsilon^2}$	$\lambda \times \Gamma$ (CUSUM, PHT); $\epsilon_{\text{cut}} \times \sqrt{\Gamma}$ (ADWIN)	scale (2nd order)	§2.3–3.5, 3.7A
ECO-L1	$\varepsilon_t/\hat{\sigma}_t$, QMLE-standardized	fitted GARCH, re-estimated per onset	location (1st order)	§4, Tab. 3
RECALIB	$(\varepsilon_t/\hat{\sigma}_t)^2$	iso-FPR bisection to 5%	scale (standardized)	§3.3
CONCEPT	$e_t^{\text{bin}} = \mathbf{1}\{\hat{y}_t \neq y_t\}$ (sign error)	exact Bernoulli$(1/2)$, no Γ estimation	location (1st order)	§3–3.7

Table 2: GARCH regimes tested and their penalty Γ (Eq. (4)); the synthetic grid fixes $\alpha = 0.08$ and solves β per target Γ. The last row applies Eq. (4) to the persistence commonly reported for daily equity indices; since Γ grows as $(1-\alpha-\beta)^{-1}$, our own variance-targeted per-asset fits sit well below it ($\hat{\Gamma} \in [2.6, 15.0]$, Section 3.7).

Regime	$\alpha + \beta$	Γ	Vol. Clustering
I.I.D. Baseline	0.00	1	None
ProteuS Cal. A	0.94–0.96	$\approx$ 4–8	Strong
ProteuS Cal. B	0.983–0.995	$\approx$ 32–110	Extreme
Synthetic grid	0.08–0.999	1–200	Controlled
S&P 500 (lit.)	$\approx$ 0.99	50–200	Extreme

Figure 3 (Appendix F) shows the consequence: both False Positive Rates (FPR) explode as Γ grows—a StrictCUSUM calibrated to a nominal 5% under IID noise fires close to or above 80% once $\Gamma > 20$; ADWIN behaves similarly. The cure is *detector-dependent*. CUSUM false alarms under H_0 follow the Siegmund-type bound $\exp(-2\delta_P\lambda/\sigma_{\text{LR}}^2)$ with long-run variance $\sigma_{\text{LR}}^2 = \Gamma$ (Appendix B), so the threshold must absorb the full inflation: $\lambda \times \Gamma$ holds the nominal level, while $\lambda \times \sqrt{\Gamma}$ leaves a residual plateau near 30% (Figure 3A). ADWIN's cut statistic compares *window means*—a standard-deviation-scale quantity—so $\epsilon_{\text{cut}} \times \sqrt{\Gamma}$ is the right correction, containing the FPR below 13% (Figure 3B). The Page–Hinkley Test (PHT), a cumulative statistic despite its running-mean reference, suffers the same explosion, needs the same $\lambda \times \Gamma$ inflation, and plateaus near 30% under $\sqrt{\Gamma}$ scaling (Section 3.5, Figure 15A).

2.4 CUSUM Delay Inflation

PROPOSITION 2.1 (CUSUM DELAY INFLATION). *Let e_t be the monitored statistic of Eq. (2) under a stationary GARCH(1,1) with $\mathbb{E}|\varepsilon_t|^{4+\delta} < \infty$ for some $\delta > 0$. To maintain the same expected time to false alarm (ARL_0), the CUSUM threshold must be recalibrated to $\lambda_{\text{GARCH}} = \lambda_{\text{iid}} \times \Gamma$, a rule exact only to leading order in λ_{iid}: the polynomial prefactor of the Siegmund ARL_0 formula, neglected here, makes it one-sidedly conservative at finite λ_{iid} and Γ (Appendix B). Under an abrupt post-change mean shift $\Delta\mu$ of the monitored statistic, the expected detection delay satisfies, to leading order in the Brownian approximation:*

$$\mathbb{E}[\text{ADD}_{\text{Data}}] = \frac{\lambda_{\text{iid}} \times \Gamma}{d} + \kappa, \tag{5}$$

where $d = \Delta\mu - \delta_P > 0$ is the effective drift magnitude, $\delta_P > 0$ is the CUSUM tolerance, and κ accounts for the stopping overshoot [33]. At fixed drift d, the delay thus grows linearly in Γ.

2.5 Scaling Law for Gradual Drifts

THEOREM 2.2 (TWO-REGIME SCALING LAW FOR GRADUAL DRIFTS). *Let the post-onset mean shift of the monitored statistic follow a linear ramp of width w, $\Delta\mu(s) = \Delta\mu_{\max} \cdot \min(s/w, 1)$, with the CUSUM recalibrated per Proposition 2.1 and run with tolerance δ_P. Write $\rho := \delta_P/\Delta\mu_{\max} < 1$ for the dead-band fraction and $w^*(\Gamma) := 2\lambda_{\text{iid}}\Gamma/[\Delta\mu_{\max}(1-\rho)^2]$ for the crossover width. In the deterministic (noise-averaged) approximation the statistic is held at zero until $s = \rho w$, then accumulates as $(\Delta\mu_{\max}/2w)(s-\rho w)^2$, so*

$$\mathbb{E}[\text{ADD}_{\text{Data}}(w,\Gamma)] \approx \begin{cases} K\sqrt{w\,\Gamma} + \rho\, w + \kappa, & w \geq w^*(\Gamma), \\ \dfrac{\lambda_{\text{iid}}\,\Gamma}{\Delta\mu_{\max} - \delta_P} + \dfrac{(1+\rho)\,w}{2} + \kappa, & w < w^*(\Gamma), \end{cases} \tag{6}$$

where $K = \sqrt{2\lambda_{\text{iid}}/\Delta\mu_{\max}}$; the branches coincide at $w = w^*(\Gamma)$, and the saturated branch reduces to Eq. (5) as $w \to 0$. The $\sqrt{w}$ term is the quadratic accumulation during the ramp; the additive ρw is the dead band, which a slow ramp must out-climb before the statistic leaves zero, and which makes the effective log-log exponent exceed $1/2$ over any bounded range of w. The inflated threshold enters as $\sqrt{\Gamma}$ in the ramp regime and as Γ in the saturated regime.

The full derivation is deferred to Appendix C.

3 The Whitening Advantage

3.1 Theoretical Statement

PROPOSITION 3.1 (SIGN-TASK WHITENING PROPERTY). *Let $\varepsilon_t = \sigma_t z_t$, where (z_t) are i.i.d., independent of the natural filtration $\mathcal{F}_{t-1}$, with a continuous distribution symmetric about zero, and $\sigma_t > 0$ is $\mathcal{F}_{t-1}$-measurable (any stationary GARCH satisfies these conditions). Consider the sign-prediction task $y_t = \mathbf{1}\{\varepsilon_t > 0\}$ and any non-anticipative classifier, i.e. any prediction $\widehat{y}_t$ measurable with respect to $\mathcal{F}_{t-1}$ (in particular an online Hoeffding Tree trained on past observations). Then the binary error stream $e_t^{\text{bin}} = \mathbf{1}\{\widehat{y}_t \neq y_t\}$ is an i.i.d. Bernoulli$(1/2)$ sequence. In particular, its autocorrelation function vanishes identically and its GARCH penalty equals 1, for every value of Γ of the input stream.*

The proof—four lines of conditioning—is given in Appendix D.

Remark 3.2 (Scope: median-type labels under conditional symmetry). The mechanism is that GARCH dependence resides in the *even* moments of the stream, from which the sign process is decoupled by conditional symmetry: binarization at the conditional median discards exactly the information where volatility clustering lives. The proposition is a property of the *task*, not of the classifier: on the stationary sign task every non-anticipative classifier—a Hoeffding Tree or a coin flip alike—has error rate exactly $1/2$, and

detection rests on the drift itself breaking the conditional symmetry. Conversely, for a non-median threshold label $y_t = \mathbf{1}\{\varepsilon_t > c\}$ with $c \neq 0$, $\mathbb{P}(y_t = 1 \mid \mathcal{F}_{t-1})$ depends on σ_t: the label stream inherits the volatility autocorrelation and the whitening guarantee fails. The two pipelines are thus complementary: a pure scale shift leaves the sign-error rate at $1/2$—invisible to Concept, exactly what the recalibrated Data pipeline catches—while a small median displacement c moves the sign-error rate *linearly* in c but the squared-residual mean only *quadratically*—the streaming analogue of the classical robustness of sign statistics to conditional heteroscedasticity [7].

Proposition 3.3 (Location/Scale Orthogonality). *Under the assumptions of Proposition 3.1, consider a post-change stream* $\varepsilon_t^\star = \mu + s\,\sigma_t z_t$ *with location shift* $\mu \in \mathbb{R}$ *and scale factor* $s > 0$. *The sign-error rate of the frozen median predictor depends on the location component only:*

$$\mathbb{P}(\varepsilon_t^\star > 0 \mid \mathcal{F}_{t-1}) = 1 - F_z\Big(-\tfrac{\mu}{s\,\sigma_t}\Big),$$

which is invariant *to* s *when* $\mu = 0$ *(a pure scale shift leaves the error rate at* $1/2$*: the Concept pipeline is exactly blind to it), and departs from* $1/2$ linearly *in* μ *at first order,* $\Delta q \simeq f_z(0)\,\mu/(s\,\sigma_t)$. *In contrast, the squared-residual mean monitored by the Data pipeline shifts* quadratically, $\mathbb{E}[(\varepsilon_t^\star)^2 \mid \mathcal{F}_{t-1}] = s^2\sigma_t^2 + \mu^2$, *carrying no first-order location signal.*

The two sensors are orthogonal: the sign sensor reads location (linear response to a median displacement, exact blindness to scale); the squared sensor reads scale (a location shift enters only at second order). This division underpins the two-branch architecture—sending a location drift to the quadratic sensor is a first-order/second-order mismatch, not a property of Γ.

Corollary 3.4 (Distribution-free calibration). *Since* (e_t^{bin}) *is i.i.d. Bernoulli*$(1/2)$ *irrespective of* Γ, *the null distribution of* any *monitoring statistic built on it—in particular a CUSUM—is universal: it does not depend on the GARCH parameters of the input stream. The Concept detector can therefore be calibrated to an exact false-alarm level through the finite Markov chain of the CUSUM recursion—or, equivalently, by direct Monte-Carlo simulation of a fair coin*—without *estimating* Γ *and* without *any asymptotic (FCLT) approximation.*

What "exact" means here. The null law is known exactly and carries no nuisance parameter, so the achieved false-alarm rate is computable to Monte-Carlo precision alone. It is not a claim that every nominal level is attainable: under a dead band δ the two-sided CUSUM increments live on a two-point lattice, so M_H is supported on multiples of 2δ and the attainable levels are discrete. At $\delta = 0.1$, $H = 5{,}000$, the levels bracketing 5% are 5.03% at $\lambda = 11.2$ and 4.29% at $\lambda = 11.4$ (2×10^5 fair-coin streams); we take the nearest attainable level at or below nominal, $\lambda^\star = 11.4$.[2] The level is granular but never unknown—the converse of the parametric route: continuous, but conditional on a fitted model.

From exact to anytime-valid. The fair-coin null licenses more than a fixed-horizon test: a mixture of betting martingales on the Bernoulli$(1/2)$ stream is a non-negative test martingale, so Ville's inequality caps its false-alarm probability at α *uniformly over all stopping times* [30, 39]. A sign-CUSUM calibrated to 5% at $H = 5{,}000$ holds there, but continued watching drives its realized false-alarm rate to 18% by $4H$; the mixture martingale stays at or below α under the same monitoring (2×10^4 fair-coin streams per level, Figure 9). The guarantee is not bought with delay: at matched false-alarm rate and moderate drift the mixture detects at least as fast as the fixed-horizon CUSUM (409 vs. 539 steps at $\eta = 0.10$), ceding ground only for abrupt shifts. Anytime-validity, which the parametric monitor cannot claim without its fitted model, thus comes for free on the whitened stream.

Proposition 3.5 (Relative Efficiency vs. Standardization). *For a location drift under Proposition 3.3, compare the sign-CUSUM (Concept) against a CUSUM on the correctly standardized residual* ε_t/σ_t *(the econometric ideal). At equal false-positive rate, the ratio of their expected detection delays* ($\mathrm{ADD}_{\mathrm{Concept}}$ *for the sign-CUSUM,* $\mathrm{ADD}_{\mathrm{Std}}$ *for the standardized one) converges to the inverse Pitman asymptotic relative efficiency of the sign test [38],*

$$\frac{\mathrm{ADD}_{\mathrm{Concept}}}{\mathrm{ADD}_{\mathrm{Std}}} \longrightarrow \frac{1}{4\,f_z(0)^2},$$

where f_z *is the standardized innovation density. This ratio exceeds* 1 *(the sign is slower) for light tails—equal to* $\pi/2 \approx 1.57$ *under Gaussianity—but drops below* 1 *(the sign is* faster*) precisely when* $f_z(0) > 1/2$, *i.e. for heavy-tailed innovations. For standardized Student-*t_ν *innovations this crossover* ($f_z(0) = 1/2$) *occurs at* $\nu \approx 4.7$.

The sign filter therefore does not trade calibration robustness for a uniform speed penalty: it trails the parametric ideal by a constant for light tails and *overtakes* it near the moment singularity (Section 4.1). Measured iso-FPR delays reproduce the prediction (Figure 4, Appendix F): the ratio runs below $1/(4f_z(0)^2)$ as $\nu \to 3$, converges to it as the tails lighten, and crosses unity at a measured $\nu^\star \approx 4.9$, above the analytic 4.7. That gap is the estimation cost: an *oracle* arm standardized by the true GARCH parameters crosses at 4.6, on the analytic prediction, so the extra 0.3 degrees of freedom is what a finite warm-up costs the parametric route.

Corollary 3.6 (Sharpe ceiling for directional drift). *Consider a directional regime whose daily returns carry an annualized Sharpe ratio* SR *under a homoscedastic Gaussian location alternative. Let* γ *be the admissible in-control average run length* (ARL_0), *and write* $\alpha_0 = 1/\gamma$. *The per-observation Kullback–Leibler divergence between the post- and pre-change return laws is* $\mathrm{KL} = \mathrm{SR}^2/504$ *(half the squared daily Sharpe, with* 252 *trading days per year; Appendix E), so the first-order lower bound of quickest detection [36] gives, in the asymptotic regime* $\gamma \to \infty$, *for* any *sequential monitor of the unconditional return law,*

$$\mathrm{ADD}_{\min} \;\geq\; \frac{\ln\gamma}{\mathrm{KL}} \;=\; \frac{504\,\ln(1/\alpha_0)}{\mathrm{SR}^2} \;\; \textit{trading days}\;\; (1 + o(1)),$$

and the corresponding floor of the sign monitor is $1/(4f_z(0)^2)$ *times larger in the small-drift limit by Proposition 3.5. Detectability of a directional regime is therefore budgeted by its Sharpe ratio alone: at the permissive* $\gamma = 20$ *used in our census, a drift of* $\mathrm{SR} = 1$ *costs at least* ≈1,510 *trading days; at the operational* $\gamma = 252$ *(one false alarm per year),* ≈2,790.

[2] The boundary case is not cosmetic: floating-point accumulation leaves M_H a few ulps above its exact lattice value, so the implemented test $M_H > \lambda^\star$ is the mathematical $M_H \geq \lambda^\star$; we report the level actually delivered.

Scope of the ceiling. The divergence $\mathrm{SR}^2/504$ uses the *unconditional* variance, so the bound is exact for a homoscedastic location alternative. Under conditional heteroscedasticity Jensen's inequality grants a variance-aware monitor a larger path divergence, $\sum_t \Delta^2/(2\sigma_t^2) \geq T\Delta^2/(2\bar{\sigma}^2)$, whereas the sign stream's budget stays the bounded Bernoulli kl. The ceiling thus budgets the binarized stream faithfully but understates likelihood monitors when a location drift meets a variance shock—the gap Section 3.7 measures.

Empirical consequences. $\mathrm{ADD}_{\mathrm{Concept}}$ is flat in Γ under a location drift (Figure 11B), and so is the 5% Concept threshold ($\lambda_C^\star \in [10.6, 10.7]$ across $\Gamma \in [1, 200]$), whereas the Data threshold must grow with Γ. The payoff is not raw speed—a correctly standardized parametric monitor is faster by the constant of Proposition 3.5—but the elimination of volatility-dependent recalibration.

3.2 Scaling Validation: Speedup vs. Γ and w

Two controlled experiments test these predictions on the Data statistic of Eq. (2) (400 seeds per configuration; Student-t_7, $\alpha = 0.08$, β solved per target Γ; the injected drift multiplies the innovation variance by s^2—a pure scale pathology—with s set per Γ so the standardized shift equals $\Delta\mu_{\max} = 2$; each Data arm calibrated to 5% false alarms by its own null quantile, against 31% under $\times\sqrt{\Gamma}$; the Concept CUSUM fixed once and for all, $\lambda_C = 10$, $\delta_C = 0.1$). Under an abrupt shift the Data delay is linear in the penalty, $\mathrm{ADD} \approx 23.7\,\Gamma + 38$, while the Concept monitor stays exactly blind, its detection rate equal to its own false-alarm level—Proposition 3.3 in action. Under gradual ramps, five penalties spanning $\Gamma \in [2, 20]$, all monitored over one common horizon so the null crossing probability is identical across Γ, reproduce Eq. (6) with no fitted constant to a median error of 5.4%; the same design prices the recalibration rule, the empirical null quantile matching $\lambda_{\mathrm{iid}} \times \Gamma$ to within 7–29%—a margin that degrades with the monitoring horizon, and whose boundary we locate at the loss of the sixth moment (Figure 5B, Appendix C).

3.3 Sensor Mismatch: Why the Squared Stream Cannot Monitor a Location Drift

Proposition 3.3 assigns each pathology to its own sensor; Table 3 (Appendix F) prices the violation. Under a location drift $\Delta = c \cdot \sigma_{\mathrm{unc}}$ (iso-FPR 5% arms), the volatility-standardized squared sensor (Recalib) runs 2 to 19× behind the two first-order monitors and its detection collapses below $c = 0.5$: the shift enters the squared stream only at second order ($\approx c^2/\sqrt{\kappa_z - 1}$), clearing the dead band $\delta_{\mathrm{R}} = 0.125$ only above $c^\star \approx 0.43$—a blind zone that persists even at $\Gamma = 1$. Eco-L1 and Concept both detect at every magnitude, their ratio governed by the Pitman efficiency of Proposition 3.5 rather than by Γ (1.66× for the parametric route at $c = 1$ under t_{30}). What separates the two routes is not delay but calibration requirements: a fitted volatility model versus a moment-free median.

3.4 Empirical Boundaries: Whiteness, Scope Limits, and Moments

We first verify the core property via Ljung–Box Q-tests [22] (lag 20) on 360 independent stationary GARCH(1,1) streams (30 seeds × 4 ETF calibrations × 3 clustering levels, $n = 8000$; online Hoeffding Tree). The squared inputs reject whiteness in 100% of the clustered calibrations ($p < 10^{-10}$) and already over-reject on the i.i.d. arm (9.2%), where t_7 innovations deprive ε_t^2 of a fourth moment and the χ^2 approximation fails; the binary errors hold the nominal level in every regime (3.3–5.0%; 4.4% pooled, 95% Wilson [2.8, 7.1]%), even through the finite-sample transient of online learning (Figure 1).

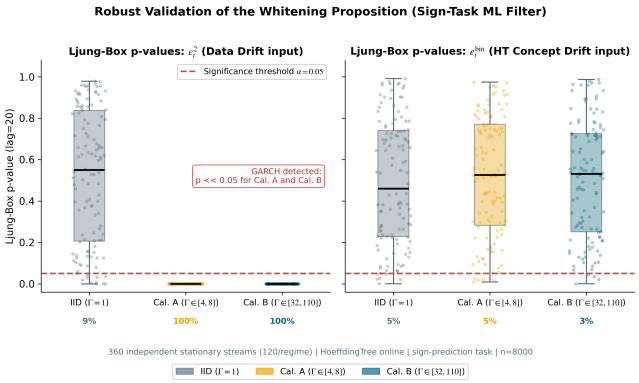


Figure 1: Proposition 3.1 in the data (Ljung–Box p-values, 360 independent stationary GARCH streams). Left: the squared returns ε_t^2 feeding the Data pipeline are overwhelmingly autocorrelated. Right: the binary classification errors e_t^{bin} show no detectable autocorrelation in any GARCH regime (3.3–5.0% rejection, the nominal type-I level).

Three stress tests then chart the boundaries (Figures 6 and 10, Appendix F). First, the guarantee needs no moments: our t_7 grid violates $\mathbb{E}[\varepsilon_t^4] < \infty$ beyond $\Gamma \approx 41.6$, yet the binary error stream stays strictly white up to $\Gamma = 200$. Second, the task boundaries are sharp: a non-median threshold or a continuous MSE loss re-inherits the autocorrelation (rejection 100% for $c \geq 0.5$ and for MSE), exactly as Remark 3.2 requires. Third, conditional asymmetry is benign but not free: skew-t innovations (realized skewness to −1.44) leave conditional independence intact yet shift the null rate to $q \approx 0.58$, making a fixed-1/2 CUSUM fire at ≈97%; recentering on a warm-up estimate $\hat{q}$ restores control. The estimate is a sample proportion, asymptotically normal with no moment requirement (Section 4).

3.5 Universality Across Detector Families

To show the advantage is a property of the signal and not of the CUSUM topology, we evaluate the Concept pipeline on four additional detectors: the Page–Hinkley Test (PHT), ADWIN, DDM [18], and EDDM [3]—a set dictated by input type, since DDM and EDDM model a Bernoulli error rate and are structurally inapplicable to the continuous Data stream.

By Corollary 3.4 the binarized stream carries an exact Bernoulli(1/2) null, so CUSUM, PHT, ADWIN, and DDM hold a bounded False Positive Rate across $\Gamma \in [1, 200]$ with no volatility-aware recalibration, and their delays under a physical shift ($c = 1.5$) are flat (Figure 15B): peak-to-peak ADD variation below 3.2% over a ×170 range of Γ for the cumulative detectors, 13% for the window-mean ADWIN. The same maximum-entropy regime exposes one design boundary: EDDM monitors the spacing between errors—Geometric(0.5) on a fair coin, extreme relative variance—keeping its degradation

ratio permanently triggered (> 90% FPR), a failure that is itself Γ-invariant: the property's signature rather than its breakdown.

The symmetric question—what monitoring the *unfiltered* stream costs—sharpens the result. Figure 11 (Appendix F) contrasts, detector by detector, the abrupt-drift delay on the raw DATA statistic against the whitened CONCEPT stream, each recalibrated as prescribed: the CONCEPT delay is flat for every detector, while the DATA delay grows at each detector's own recalibration exponent (log-log slopes 0.86 CUSUM, 1.09 PHT, 0.47 ADWIN). We call this the *Lethargy Tax*: controlling false alarms on a heteroscedastic stream defers detection by the very factor used to inflate the threshold. The running-mean PHT pays it twice—beyond $\Gamma \approx 75$ its adaptive reference absorbs the shift faster than the inflated threshold reacts, collapsing detection below 50% (a *stochastic syncope*). The whitening filter voids the entire schedule of penalties.

3.6 The Conditional Mean Boundary (AR-GARCH)

Proposition 3.1 requires the binarization threshold to sit at the conditional median. Under a non-zero conditional mean $\mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$—e.g. AR(1) momentum, $\mu_t = \phi r_{t-1}$—the naive sign $y_t^{\text{naive}} = \mathbf{1}\{r_t > 0\}$ has $\mathbb{P}(r_t > 0 \mid \mathcal{F}_{t-1}) = 1 - F_z(-\mu_t/\sigma_t)$, which couples with σ_t and reinstates the GARCH autocorrelation in the label stream. The filter therefore applies to the *residuals* of the conditional-mean model; the practical question is what survives when those residuals are formed with an *estimated* $\hat{\mu}_t$.

Estimation enters through the bias, not the variance. The proposition asks of the recentering only measurability, not accuracy. Any $\mathcal{F}_{t-1}$-measurable $\hat{\mu}_t$ leaves the sign conditionally Bernoulli with parameter $q_t = 1 - F_z(u_t)$, where $u_t := (\hat{\mu}_t - \mu_t)/\sigma_t$ is the centering error in innovation units. Symmetry of f_z annihilates the quadratic term, so $q_t - 1/2 = -f_z(0)\, u_t + O(u_t^3)$ and $\mathrm{Cov}(y_t, y_{t+k}) \simeq f_z(0)^2 \,\mathrm{Cov}(u_t, u_{t+k})$ for $k \geq 1$: the *systematic* component of u_t perturbs the label stream at first order, its zero-mean component only at second order. The prediction is falsifiable, and asymmetric: Ljung–Box responds to $|\mathrm{Cov}(y_t, y_{t+k})|$ whereas the CUSUM responds to its sign, so a systematic error of magnitude b should cost the same whiteness whether it under- or over-centers, but should inflate the false-alarm rate only when it under-centers.

Measurement. We test it on AR(1)-GARCH(1,1) streams ($\alpha = 0.1058$, $\beta = 0.8742$, standardized t_7; 10,000 paired trajectories per ϕ, horizon $H = 5{,}000$), recentering with a non-anticipative rolling OLS of window n against naive ($\hat{\mu}_t = 0$) and oracle ($\hat{\mu}_t = \phi r_{t-1}$) arms on the same paths (Figure 7). The naive boundary is a slope, not a cliff: Ljung–Box rejection climbs from 5.1% at $\phi = 0$ to 99.8% at $\phi = 0.15$, dragging the CONCEPT FPR to 20.8%. Every OLS arm erases that failure: across the full 7×4 grid of momentum strengths $\phi \in [0, 0.15]$ and window lengths $n \in \{125, 250, 500, 1000\}$, Ljung–Box rejection stays within 4.6–5.6% and FPR within 4.3–5.9%, matching the oracle bands on the same paths. The two channels separate as predicted: the systematic one, the classical small-sample AR bias $\mathbb{E}[\hat{\phi}] - \phi \approx -2.5\,\phi/n$, stays under 2.9×10^{-3}, while the dispersion channel, whose RMSE reaches 11.4% of σ_{unc} at $n = 125$, costs at most 0.4 points of rejection. Bias, not noise, is what the property is exposed to, and this holds on a DGP past the fourth-moment boundary ($\mathbb{E}[(\alpha z_t^2 + \beta)^2] = 1.005 > 1$), where Γ is formal only and the recentered sign stream is white regardless. These are H_0 measurements: they bound the *calibration* cost of estimation, not its delay cost.

The adverse direction. We test the asymmetry directly by injecting a deliberate bias into an otherwise unbiased arm—$\hat{\mu}_t = (\hat{\phi}_t + b)\, r_{t-1}$ on a stream with $\phi = 0$, $n = 250$, 10,000 trajectories per b (Figure 8). Whiteness behaves exactly as predicted: the injected-bias arm and the naive arm at $\phi = b$ reject within three points of each other across a range spanning 5 to 100%. Calibration does not, and the asymmetry is the whole point: injected *over*-centering makes the sign stream negatively autocorrelated and the false-alarm rate collapses to 0.86% at $b = 0.15$, while naive *under*-centering inflates it to 20.8%—identical |Cov|, opposite sign, opposite consequence. The rolling OLS errs on the adverse side (its bias leaves a residual momentum $+2.5\,\phi/n > 0$, so it under-centers), but its safety rests on magnitude, and the margin is wide: at a residual momentum of 0.02—seven times the largest we measure—the under-centering penalty is still only 1.1 points of false-alarm rate.

3.7 Real-World Validation: FirstRate ETF Data In-The-Wild

To test whether the advantage survives market microstructure and empirical heavy tails, we move to four reference ETFs (SPY, PFF, VNQ, BWX) from FirstRate Data[3], downsampled to daily log-returns of 16:00 ET closes (splits and dividends adjusted) to preserve the Martingale Difference Sequence hypothesis that intraday bid-ask bounce and U-shaped seasonality violate.

Warm-up and Gatekeeper Validation (pre-2020 window). On the pre-pandemic window we recenter the binary error stream on the empirical conditional median; a lag-20 Ljung–Box test finds no serial correlation on any asset (SPY $p = 0.22$, PFF 0.57, VNQ 0.96, BWX 0.77), licensing the filter. The DATA pipeline needs more machinery: variance-targeting QMLE on the centered warm-up innovations yields markedly heterogeneous per-asset penalties ($\hat{\Gamma} =$ 15.0, 2.6, 4.2, 5.8 for SPY, PFF, VNQ, BWX).[4] The sign filter needs none of this.

Panel A: The COVID-19 Variance Shock (2020). We monitor SPY through 2020 (Figure 2A, Appendix F), tracking S_t/λ for the recalibrated DATA monitor ($\lambda_{\text{iid}} = 65$, $\delta_P = 0.5$, one-sided) and the whitened CONCEPT monitor ($\lambda_C = 10$, $\delta_C = 0.1$, two-sided). Neither fires—DATA peaks at 0.37 of its threshold, CONCEPT at 0.45—but the silences differ in kind. The DATA threshold, inflated to $\lambda_{\text{iid}} \times \hat{\Gamma}_{\text{SPY}} \approx 975$ by the fitted persistence, is precisely the barrier that defers even a legitimate variance alarm: the Lethargy Tax on real data. The CONCEPT monitor is silent because binarization caps the Fisher information below what an alarm requires within the crash's short duration—a structural limit quantified below. One

[3] https://firstratedata.com, product `full_1min_adjsplitdiv`. The commercial license forbids redistribution of raw prices; the full pipeline code and all derived alarm and summary CSVs are released at https://github.com/TheWhiteningAdvantage-KDD2027/The-Whitening-Advantage-Experiments (Appendix A).

[4] A naive QMLE on the raw, unscaled returns silently returns its initialization on these micro-variance streams (a floating-point conditioning failure with `success=True`); variance targeting—certified by a parameter-recovery test on simulated ground truth—is what makes the parametric baseline usable here at all.

pipeline misses the event it was built to catch; the other runs out of budget for an event too brief to afford.

Panel B: Semi-Real Injection and Statistical Degradation (2021–2023). For quantitative delay we inject a permanent directional shift ($\Delta = 1.5\sigma_{\text{unc}}$, a deliberately aggressive stress probe) into the real 2021–2023 streams, resetting the monitors on the first trading day of each month (36 onsets per ETF, 250-day horizon). Both pipelines watch the same injected series, each recalibrated at every onset from the trailing 250 days (the DATA arm refitting $\hat{\Gamma}$ by QMLE at each onset). The injection is synthetic because an exact ground-truth onset requires it; the background noise is strictly empirical, and a *placebo arm* ($\Delta = 0$) measures the alarm rate induced by true variance regimes such as the 2022 bear market.

The raw DATA pipeline shows only weak, placebo-confounded power against this location drift, exactly as Proposition 3.3 predicts for a squared sensor: raw detection never exceeds one third of onsets (SPY 0.0%; PFF 30.6% against a 22.2% placebo; VNQ 16.7%; BWX 19.4%)—the alarms that do occur respond to the second-order term $(\Delta/\sigma)^2$, confounded with genuine variance regimes. The whitened CONCEPT pipeline, monitoring the correct first-order signal, separates cleanly: 100% detection on all four ETFs at an ADD of 36.6–64.6 days, against background alarm rates of 0–13.9%. The magnitude implies a Sharpe ratio far beyond any traded index; the experiment calibrates *power*, and its message is architectural—only one of the two sensors responds to a directional break at all.

The Regime Census and the Sharpe Ceiling In-The-Wild. The natural question is what *unforced* directional regimes the market offers. A retrospective multi-scale Pagan–Sossounov bull/bear dating [28] of the four streams (2000–2025; 66 phases after duration censoring, the COVID-19 crash—too brief for the filter—dated at the raw scale) answers through Corollary 3.6: for 53 of 66 phases (80%), the floor $\text{ADD}_{\min}$ at the most permissive operating point exceeds the phase's own duration. Pricing the binarization exactly through $\text{kl}(q_{\text{phase}} \,\|\, q_{\text{ref}})$ moves that count by one phase (52 of 66); tightening to one false alarm per year leaves 64 of 66 (97%) out of budget. Long secular advances move the up-day probability only marginally (SPY 2011–2018: 0.541 → 0.554 over 1,753 days); they nonetheless dominate the detectable set on duration alone, and even there the floor consumes 55–92% of the phase. What falls out of budget is the short episode—duration binds, not magnitude.

The COVID-19 crash ($\Delta q \approx -0.28$, annualized Sharpe ≈ -6.0 over 23 trading days) separates the two readings of the ceiling. For the *sign* stream the budget is the exact Bernoulli divergence $\text{kl}(q_{\text{phase}} \,\|\, q_{\text{ref}}) = 0.162$ nats/day: a floor of ≈34 trading days at one false alarm per year, and 18.5 days at the loosest calibration—four fifths of the phase. A variance-aware monitor is not so bounded: in a look-ahead oracle backtest (Figure 14, Appendix F), a CUSUM on returns standardized by the conditional volatility of a GARCH(1,1) fitted on a window *including* the crash—parameter oracle, causal filtration—detects it in 3 trading days (likelihood-ratio increments, phase false-alarm probability 1.3%) to 16 days (standardized-mean CUSUM), against a bootstrap null freezing the same volatility path. The mechanism is Jensen's inequality: the path divergence $\sum_t \Delta^2/(2\sigma_t^2)$ is 10.6× the unconditional budget, because the conditional σ_t lags the shock. The same protocol discriminates the census flags (2009 recovery detected, 2019 advance missed, no alarm on the 2011 correction at the matched operating point), and the detection is specific to *conditional* information: a look-ahead *centered realized* volatility, pricing the crash into σ_t contemporaneously, yields no alarm at iso-FPR. The 3-day figure is an upper envelope no online monitor attains.

4 Discussion

4.1 Econometric Whitening: Baseline, Misspecification, and the Moment Singularity

The whitening filter is conceptually adjacent to GARCH-residual standardization [21, 34]. Under a correctly specified process and a converged QMLE fit, monitoring ECO-L1 also achieves a Γ-invariant false-alarm rate with flat delays and, for a location drift, the speed advantage of Proposition 3.5 (Table 3): continuous residuals carry more information per observation than a binarized stream. The sign filter's value is therefore not speed under correct specification but the removal of the specification and estimation steps—two costs the parametric route pays and the sign filter does not:

Estimation cost. ECO-L1 must fit a persistent GARCH ($\alpha + \beta = 0.85$) on a finite warm-up: on a 250-step window the estimated persistence collapses to a median $\hat{\alpha} + \hat{\beta} = 0.62$ and the FPR nearly doubles to 9.5%; the level is restored from $n = 500$ onward. The asymmetry is structural: the sample median is asymptotically $\mathcal{N}(0,\ 1/(4nf_z(0)^2))$ with *no* finite-moment requirement [38], while QMLE normality requires a finite fourth moment ($\kappa_z < \infty$) and its variance explodes under financial heavy tails. The sign pipeline is warm-up-independent in practice (measured FPR 3–8% across all warm-up lengths).

Efficiency reversal. Figure 4B measures $\text{ADD}_{\text{Concept}}/\text{ADD}_{\text{Eco-L1}}$ against Student-t_ν degrees of freedom at matched 5% FPR: the ratio follows the qualitative shape of $1/(4f_z(0)^2)$ and inverts in the heavy-tailed regime of the moment singularity (Section 4.1), while never exceeding the Gaussian ceiling $\pi/2$ elsewhere.

Efficiency reversal on a real heavy-tailed stream. We race the recentred sign-CUSUM (CONCEPT) against a CUSUM on the honestly standardized residual $\varepsilon_t/\hat{\sigma}_t$ (ECO-L1; GARCH(1,1) refitted on each pre-onset window and filtered forward through the *observed* stream—no look-ahead) on daily Bitcoin returns (2015–2025; 106 monthly onsets, public adjusted closes), both arms at an identical 4.7% false-alarm rate on real placebo windows. On BTC, whose standardized innovations carry $\hat{\nu} = 2.78$—well inside the heavy-tailed regime of Proposition 3.5—the sign filter leads across the reliable range: the delay ratio runs from 0.74 at $c = 0.35$ to parity (1.01) at $c = 1.5$, mean 0.87. A quasi-Gaussian synthetic control (t_{30}, identical onsets and false-alarm rate) inverts the ordering to ECO-L1-faster (0.98–1.14, mean 1.06), reproducing the light-tailed Pitman penalty on real machinery (Figure 16, Appendix F). Ethereum marks the practical boundary: its recentred sign stream fails whiteness (Ljung–Box $p = 0.019$), the fair-coin pivot does not hold exactly, and the synthetic control does not recover the light-tailed ordering at its 72 onsets; we report BTC as the demonstration and ETH as the frontier.

Misspecification. Beyond finite-sample instability, explicit standardization is exposed to structural misspecification. When the

stream carries asymmetric volatility clustering (GJR-GARCH, leverage $\gamma_{\text{lev}} > 0$) but the baseline standardizes with its symmetric GARCH(1,1) *population limit*—so any leakage is structural rather than estimation noise—residual Ljung–Box rejection on the standardized squared residual rises from 5.1% to 24.6% at $\gamma_{\text{lev}} = 0.28$ (Figure 12, Appendix F), driving its false-alarm rate from 3.2% to 20.6% under a pseudo-Gaussian innovation law that isolates leverage from tail effects. The sign pipeline is structurally immune to leverage itself: σ_t depends asymmetrically on the sign of ε_{t-1} but remains $\mathcal{F}_{t-1}$-measurable, so Proposition 3.1 holds exactly: across the whole leverage grid its Ljung–Box rejection stays within 4.6–5.4% and its false-alarm rate within 7.6–8.4% at its fixed threshold—above nominal, but flat where the baseline's climbs by a factor of six. The parametric route therefore fails to control false alarms even at the population limit once misspecified; the whitened route holds a leverage-invariant rate throughout.

The moment singularity. Standard approximations, Γ included, require $\mathbb{E}[\varepsilon_t^4] < \infty$. Stressing both pipelines as Student-t innovations approach the kurtosis singularity ($\nu \to 4$; drift $c = 1.0$), the Data pipeline sinks into a *stochastic syncope* as the sample variance of the squared residuals diverges: detection decays monotonically (83% at $\nu = 10$, 61% at $\nu = 7$), collapses below the 50% censoring threshold for $\nu \le 5.5$, and the surviving minority carries survivorship-biased delays of 2,400–3,000 steps (Figure 13, Appendix F). The Concept pipeline, reading only the topology of the conditional median, stays flat at 34–38 steps right up to the singularity.

When monitoring continuous regression outputs (e.g., deep-learning forecasters), the binarization filter is absent and the continuous loss streams may re-inherit the GARCH autocorrelation—a natural extension for monitoring forecasting pipelines.

Operating regimes where the budget is affordable. Corollary 3.6 prices the ceiling *per observation*, so the calendar-time floor of a location drift is fixed by the annualized Sharpe alone, invariant to sampling frequency. Daily equities are the adversarial case (SR = 1 is $\text{SR}_{\text{daily}} \approx 0.063$), which is why 80% of dated regimes fall out of budget. Exact calibration pays off elsewhere: on large-shift heteroscedastic streams (network, sensor, industrial telemetry), where shifts far above the noise floor clear the Bernoulli budget in a handful of observations; and on long-horizon slow drifts, where duration is not binding and anytime-validity removes the horizon-dependence of the alarm rate. The guarantee transfers verbatim, but this paper delimits these regimes without measuring them.

5 Related Work

Sequential change detectors—CUSUM [29], ADWIN [5]—and the concept-drift literature they anchor [19, 23] are calibrated under i.i.d. or weakly dependent noise. Recent benchmarks document detector-calibration fragility on real streams and the evaluation distortions temporal dependence induces [24, 40]; financial evaluations of DDM/EDDM/ADWIN report calibration difficulties on market data [9, 26]; none models the volatility-clustering mechanism itself. Change-point analysis under dependence spans self-normalization [31] and retrospective GARCH-monitoring statistics [2, 4, 21, 34]; all presuppose a fitted parametric model—exactly the failure mode quantified in Section 4.

Our whitening result is a statement about the null: under H_0 the monitored stream is an *exact pivot*—a fair coin—whatever the nuisance dynamics. Probability-integral-transform diagnostics [14] reach i.i.d. uniformity only under a correctly specified conditional law. Sign statistics are classically robust to conditional heteroscedasticity [7], and sign predictability under volatility dynamics is documented [12]; Corollary 3.4 turns these into a calibration recipe with no specification step. Because the pivot is exact, a test martingale on the fair coin inherits the time-uniform false-alarm control of e-processes [30, 32, 39]—unavailable to monitors that condition on a fitted volatility model. For evaluation we favor a semi-synthetic protocol with exact onsets and interpretable Γ [35] over deep generative simulators [10]; continuous-time quickest detection [11] targets a different observation model.

6 Conclusion

We have established the **Whitening Advantage**: on sign-prediction tasks under conditionally symmetric innovations, binarization by a non-adaptive classifier yields an i.i.d. Bernoulli(1/2) error stream (Proposition 3.1), making Concept-drift detection structurally insensitive to volatility clustering. On the Data side we derived the penalty factor Γ and its detector-dependent recalibrations; the two sensors are orthogonal (Proposition 3.3). At matched false-positive rate the sign filter trails correctly specified parametric standardization by $1/(4f_z(0)^2)$ (Proposition 3.5), inverting below a measured $\nu^\star \approx 4.9$ (analytic 4.7 for the exactly standardized ideal). Its decisive advantage is calibration, not delay: the sign stream is a fair coin by construction, exact in finite samples with no volatility model, where the parametric approach is conditional on a converged fit and a sufficient warm-up.

On real daily ETF streams (FirstRate Data, 2000–2025), neither monitor fires during the 2020 COVID-19 shock: the raw monitor is throttled by its Γ-inflated threshold, while the crash's brief duration exhausts the sign filter's information budget. Under injected breaks the picture inverts, the sign filter detecting on every asset. A retrospective census explains why unforced drifts are hard: the Sharpe-ceiling floor exceeds the duration of 80% of dated bull/bear episodes. A look-ahead oracle locates the boundary: a variance-aware monitor granted look-ahead GARCH parameters detects the crash in 3–16 days—a 10.6× Jensen gain no online estimator attains. Real directional drift is detectable in principle but evanescent in practice; exactly why exact, variance-estimation-free calibration matters more than raw speed.

Escaping the ceiling sideways. The ceiling is univariate; pooling K correlated streams is a real but small escape (97 surviving US equities, 2005–2025, public adjusted closes—a survivorship-biased panel providing an upper-bound co-movement benchmark; Figure 17): sign correlation $\hat{\rho} \approx 0.26$ saturates the effective panel near $1/\hat{\rho} \approx 3.8$, the budget contracts by a measured 2× only, and temporal whiteness fails beyond $K = 10$—an independence null reaches ∼100% false alarms by $K = 40$, forcing bootstrap calibration. The pooled monitor still never flags the 2020 crash: pooling widens the target, it does not move the wall.

References

[1] Elena Andreou and Eric Ghysels. 2002. Detecting multiple breaks in financial market volatility dynamics. *Journal of Applied Econometrics* 17, 5 (2002), 579–600. doi:10.1002/jae.684

[2] Alexander Aue and Lajos Horváth. 2013. Structural breaks in time series. *Journal of Time Series Analysis* 34, 1 (2013), 1–16. doi:10.1111/j.1467-9892.2012.00819.x

[3] Manuel Baena-García, José del Campo-Ávila, Raúl Fidalgo, Albert Bifet, Ricard Gavaldà, and Rafael Morales-Bueno. 2006. Early drift detection method. In *Fourth International Workshop on Knowledge Discovery from Data Streams*. 77–86.

[4] István Berkes, Edit Gombay, Lajos Horváth, and Piotr Kokoszka. 2004. Sequential change-point detection in GARCH(p,q) models. *Econometric Theory* 20, 6 (2004), 1140–1167. doi:10.1017/S0266466604206041

[5] Albert Bifet and Ricard Gavaldà. 2007. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*. 443–448. doi:10.1137/1.9781611972771.42

[6] Tim Bollerslev. 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31, 3 (1986), 307–327. doi:10.1016/0304-4076(86)90063-1

[7] Bryan Campbell and Jean-Marie Dufour. 1995. Exact nonparametric orthogonality and random walk tests. *The Review of Economics and Statistics* 77, 1 (1995), 1–16. doi:10.2307/2109988

[8] Marine Carrasco and Xiaohong Chen. 2002. Mixing and moment properties of various GARCH and stochastic volatility models. *Econometric Theory* 18, 1 (2002), 17–39. doi:10.1017/S0266466602181023

[9] Rodolfo C. Cavalcante and Adriano L. I. Oliveira. 2015. An approach to handle concept drift in financial time series based on Extreme Learning Machines and explicit Drift Detection. In *2015 International Joint Conference on Neural Networks (IJCNN)*. 1–8. doi:10.1109/IJCNN.2015.7280721

[10] Adil Rengim Cetingoz and Charles-Albert Lehalle. 2025. Synthetic data for portfolios: A throw of the dice will never abolish chance. arXiv preprint arXiv:2501.03993. doi:10.48550/arXiv.2501.03993

[11] Etienne Chevalier, Yadh Hafsi, and Vathana Ly Vath. 2023. Uncovering market disorder and liquidity trends detection. arXiv preprint arXiv:2310.09273.

[12] Peter F. Christoffersen and Francis X. Diebold. 2006. Financial Asset Returns, Direction-of-Change Forecasting, and Volatility Dynamics. *Management Science* 52, 8 (2006), 1273–1287. doi:10.1287/mnsc.1060.0520

[13] Rama Cont. 2001. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance* 1, 2 (2001), 223–236. doi:10.1080/713665670

[14] Francis X. Diebold, Todd A. Gunther, and Anthony S. Tay. 1998. Evaluating density forecasts with applications to financial risk management. *International Economic Review* 39, 4 (1998), 863–883. doi:10.2307/2527342

[15] Zhuanxin Ding, Clive W. J. Granger, and Robert F. Engle. 1993. A long memory property of stock market returns and a new model. *Journal of Empirical Finance* 1, 1 (1993), 83–106. doi:10.1016/0927-5398(93)90006-D

[16] Robert F. Engle. 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50, 4 (1982), 987–1007. doi:10.2307/1912773

[17] Christian Francq and Jean-Michel Zakoïan. 2010. *GARCH Models: Structure, Statistical Inference and Financial Applications*. Wiley.

[18] João Gama, Pedro Medas, Gladys Castillo, and Pedro Pereira Rodrigues. 2004. Learning with drift detection. In *Advances in Artificial Intelligence – SBIA 2004*. Springer Berlin Heidelberg, 286–295. doi:10.1007/978-3-540-28645-5_29

[19] João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. 2014. A survey on concept drift adaptation. *Comput. Surveys* 46, 4 (2014), 1–37. doi:10.1145/2523813

[20] Norbert Herrndorf. 1984. A functional central limit theorem for weakly dependent sequences of random variables. *The Annals of Probability* 12, 1 (1984), 141–153. doi:10.1214/aop/1176993379

[21] Lajos Horváth, Piotr Kokoszka, and Aonan Zhang. 2006. Monitoring constancy of variance in conditionally heteroskedastic time series. *Econometric Theory* 22, 3 (2006), 373–402. doi:10.1017/S0266466606060191

[22] G. M. Ljung and G. E. P. Box. 1978. On a measure of lack of fit in time series models. *Biometrika* 65, 2 (1978), 297–303. doi:10.1093/biomet/65.2.297

[23] Jie Lu, Anjin Liu, Fan Dong, Feng Gu, João Gama, and Guangquan Zhang. 2018. Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering* 31, 12 (2018), 2346–2363. doi:10.1109/TKDE.2018.2876857

[24] Daniel Lukats, Oliver Zielinski, Axel Hahn, and Frederic Stahl. 2025. A benchmark and survey of fully unsupervised concept drift detectors on real-world data streams. *International Journal of Data Science and Analytics* 19, 1 (2025), 1–31. doi:10.1007/s41060-024-00620-y

[25] Florence Merlevède, Magda Peligrad, and Sergey Utev. 2006. Recent advances in invariance principles for stationary sequences. *Probability Surveys* 3 (2006), 1–36. doi:10.1214/154957806100000202

[26] Filippo Neri. 2021. Domain Specific Concept Drift Detectors for Predicting Financial Time Series. arXiv preprint arXiv:2103.14079. doi:10.48550/arXiv.2103.14079

[27] Whitney K. Newey and Kenneth D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55, 3 (1987), 703–708. doi:10.2307/1913610

[28] Adrian R. Pagan and Kirill A. Sossounov. 2003. A simple framework for analysing bull and bear markets. *Journal of Applied Econometrics* 18, 1 (2003), 23–46. doi:10.1002/jae.664

[29] E. S. Page. 1954. Continuous inspection schemes. *Biometrika* 41, 1/2 (1954), 100–115. doi:10.1093/biomet/41.1-2.100

[30] Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. 2023. Game-Theoretic Statistics and Safe Anytime-Valid Inference. *Statist. Sci.* 38, 4 (2023), 576–601. doi:10.1214/23-STS894

[31] Xiaofeng Shao and Xianyang Zhang. 2010. Testing for change points in time series. *J. Amer. Statist. Assoc.* 105, 491 (2010), 1228–1240. doi:10.1198/jasa.2010.tm10103

[32] Jaehyeok Shin, Aaditya Ramdas, and Alessandro Rinaldo. 2024. E-detectors: A Nonparametric Framework for Sequential Change Detection. *The New England Journal of Statistics in Data Science* 2, 2 (2024), 229–260. doi:10.51387/23-NEJSDS51

[33] David Siegmund. 1985. *Sequential Analysis: Tests and Confidence Intervals*. Springer-Verlag.

[34] Junmo Song and Jiwon Kang. 2020. Sequential change point detection in ARMA-GARCH models. *Journal of Statistical Computation and Simulation* 90, 8 (2020), 1520–1538. doi:10.1080/00949655.2020.1734807

[35] Andrés L. Suárez-Cetrulo, Alejandro Cervantes, and David Quintana. 2025. ProteuS: A Generative Approach for Simulating Concept Drift in Financial Markets. *arXiv preprint arXiv:2509.11844* (2025). doi:10.48550/arXiv.2509.11844

[36] Alexander G. Tartakovsky, Igor V. Nikiforov, and Michèle Basseville. 2014. *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Chapman and Hall/CRC.

[37] Ruey S. Tsay. 2010. *Analysis of Financial Time Series* (3rd ed.). Wiley.

[38] A. W. van der Vaart. 1998. *Asymptotic Statistics*. Cambridge University Press, Cambridge.

[39] Jean Ville. 1939. *Étude critique de la notion de collectif*. Gauthier-Villars, Paris.

[40] Giacomo Ziffer, Federico Giannini, and Emanuele Della Valle. 2024. TeNet: Benchmarking Data Stream Classifiers in Presence of Temporal Dependence. In *2024 IEEE International Conference on Big Data (BigData)*. 1187–1196.

A Reproducibility

Code, data, and provenance. All code, protocol CSVs, and reproduction scripts are available anonymously at https://github.com/TheWhiteningAdvantage-KDD2027/The-Whitening-Advantage-Experiments. The repository includes all derived CSVs for FirstRate ETFs (redistribution of raw prices restricted), public fetchers for Bitcoin, Ethereum, and 97 equities, and a complete mapping table in `README.md` linking every figure and number to its generating script and CSV.

Compute and implementation. Experiments ran on an AMD EPYC 8224P CPU (24 cores, 192 GB RAM), without GPU. Regenerating every figure and number reported here from cached inputs takes ≈1 h of wall-clock time, of which ≈25 min is the iso-FPR race of Section 3.3; acquiring the public equity panel and cryptocurrency series is a further one-off ≈2 h bounded by provider rate limits rather than by computation. The repository also ships a higher-resolution scaling campaign (≈45 min) that the paper does not use. Simulations parallelize over seeds via `joblib` and `concurrent.futures`; bit-level reproducibility requires single-threaded BLAS, which we enforce and which costs wall-clock time. QMLE recursions and Fernández–Steel skew-t distributions are implemented natively in the repository to avoid third-party package defaults (`arch`). Regime phases are cut *post-onset*: the return dated at a turning point closes the regime it ends, so consecutive phases partition the return series and no pre-change observation enters a phase's Sharpe. The convention matters: at troughs following a crash the turning-point return is an outlier (e.g. −18.6% on PFF, 2020-03-18) that both depresses the mean and inflates the variance of the phase that follows, biasing its floor upward. Scaling-law campaigns run under two horizons

($H = 2 \times 10^5$ for Section 3.2 and $H = 3 \times 10^6$ for Appendix C's boundary test).

B Proof of Proposition 2.1

Proof. Under GARCH(1,1) stationarity, (ε_t) is geometrically β-mixing [8], hence so is the transformed sequence (e_t) of Eq. (2). Under $\mathbb{E}|\varepsilon_t|^{4+\delta} < \infty$, the FCLT for strongly mixing sequences applies [20, 25]: the partial sums of $\xi_t := e_t - \delta_P$ satisfy an invariance principle with long-run variance $\sigma^2_{\mathrm{LR}} = \Gamma$ (unit marginal variance after standardization), where $\Gamma = 1 + 2\sum_{k=1}^{\infty} \rho_e(k)$ is the normalized spectral density at frequency zero [27].

Threshold inflation. In the Brownian approximation, the CUSUM statistic under H_0 is a reflected Brownian motion with drift $-\delta_P$ and diffusion coefficient σ^2_{LR}, whose expected first-passage time over λ is

$$\mathrm{ARL}_0(\lambda) = \frac{\sigma^2_{\mathrm{LR}}}{2\delta_P^2}\left(e^{2\delta_P\lambda/\sigma^2_{\mathrm{LR}}} - \frac{2\delta_P\lambda}{\sigma^2_{\mathrm{LR}}} - 1\right),$$

dominated by the exponential term for large λ [33, 36]. Holding ARL_0 fixed when σ^2_{LR} inflates from 1 to Γ requires holding the exponent $2\delta_P\lambda/\sigma^2_{\mathrm{LR}}$ fixed up to logarithmic terms, i.e. $\lambda_{\mathrm{GARCH}} = \lambda_{\mathrm{iid}} \times \Gamma$ asymptotically. The neglected polynomial prefactor leaves a finite-Γ error that is bounded and one-sided on our grid: against a 5% nominal level, the realized FPR under the rule falls from 2.7% at $\Gamma = 1.2$ to 1.3% at $\Gamma = 200$, monotonically and never in excess (Figure 3A). The rule is therefore safe but increasingly costly in power as Γ grows—the Lethargy Tax of Section 3.5 in its calibration form.

Delay. Under the alternative, the increments have positive mean $d = \Delta\mu - \delta_P$; by Wald's identity applied to the stopped accumulation (valid to first order for ergodic increments), $\mathbb{E}[S_\tau] = \lambda_{\mathrm{GARCH}} + \mathbb{E}[\text{overshoot}]$, whence

$$\mathbb{E}[\mathrm{ADD}] \approx \frac{\lambda_{\mathrm{iid}} \times \Gamma}{d} + \kappa,$$

with κ the overshoot correction [33]; the delay is thus linear in Γ at fixed drift. □

C Proof of Theorem 2.2

Proof. The expected CUSUM increment at step s after drift onset is $\mathbb{E}[\xi_{\tau^*+s}] = \Delta\mu_{\max}\min(s/w, 1) - \delta_P$, which is *negative* for $s < \rho w$: the reflection at zero holds the mean statistic there, and accumulation starts at $s_0 = \rho w$. Integrating from s_0 and completing the square,

$$S_{\tau^*+s} \approx \begin{cases} \dfrac{\Delta\mu_{\max}}{2w}(s - \rho w)^2, & s \le w, \\ \dfrac{\Delta\mu_{\max}(1-\rho)^2 w}{2} + (\Delta\mu_{\max} - \delta_P)(s - w), & s > w. \end{cases}$$

Ramp regime. Solving $\frac{\Delta\mu_{\max}}{2w}(s^* - \rho w)^2 = \lambda_{\mathrm{iid}}\Gamma$ gives $s^* = K\sqrt{w\Gamma} + \rho w$; the condition $s^* \le w$ is equivalent to $w \ge w^*(\Gamma) = 2\lambda_{\mathrm{iid}}\Gamma/[\Delta\mu_{\max}(1-\rho)^2]$.

Saturated regime. Otherwise the threshold is crossed after ramp completion: the linear branch gives $s^* = \lambda_{\mathrm{iid}}\Gamma/(\Delta\mu_{\max} - \delta_P) + (1+\rho)w/2$, which is Eq. (5) at $w = 0$. Both expressions equal w^* at $w = w^*(\Gamma)$.

Adding the irreducible overshoot correction κ [33] completes the proof.

Reading of the dead band. The tolerance δ_P is the detector's reference drift: it protects the false-alarm rate by refusing to accumulate anything smaller. Against a ramp, the instantaneous shift enters at zero and must first climb past that floor, so the detector is structurally blind over the first fraction ρ of the ramp and only then begins to integrate. The blind window is ρw—proportional to the drift's own width, not to the threshold, not to Γ. This is a property of every CUSUM run with a non-zero tolerance, not an artefact of the deterministic approximation, and it inverts the usual intuition about slow drifts: since $K\sqrt{w\Gamma}$ grows as $\sqrt{w}$ while ρw grows linearly, the dead band overtakes the accumulation term at $w = w^*(\Gamma)(1-\rho)^2/\rho^2$ and dominates beyond it, so

$$\mathrm{ADD}_{\mathrm{Data}}(w, \Gamma) \sim \frac{\delta_P}{\Delta\mu_{\max}} w \qquad (w \to \infty),$$

a delay linear in the ramp width and *independent of both the GARCH penalty and the threshold.* Arbitrarily slow drifts are therefore not penalized by heteroscedasticity at all; they are penalized by the detector's own tolerance. Our grid reaches 2.5× that turning point, where ρw already contributes 58% of the predicted delay.

Measured domain of validity. We test Eq. (6) on five penalties, $\Gamma \in \{2, 4, 8, 11.6, 20\}$, with w swept on a grid held fixed in units of $w^*(\Gamma)$, from deep inside the saturated regime to $22.5\,w^*$, and *all* penalties monitored over one common horizon ($H = 200{,}000$) so that the null crossing probability is identical across Γ and the thresholds are comparable in level (censoring $\le 1.3\%$, detection $\ge 98.7\%$ everywhere; realized hold-out false-alarm rate 3.5–6.2%).

That design first prices the recalibration rule itself. The empirical null quantile satisfies $\lambda^\star(\Gamma)/\Gamma \in [138, 167]$ against $\lambda_{\mathrm{iid}} = 129.5$ measured at the same horizon—the $\lambda_{\mathrm{iid}} \times \Gamma$ rule of Proposition 2.1 holding to within 7–29% over a tenfold range of Γ, with the residual on the conservative side.

Setting $\kappa = 0$, Eq. (6) then predicts the measured delay with *no* fitted constant to a median error of 5.4% (maximum 27%). Dropping the ρw term—the form in which the deterministic limit is usually stated—leaves a prediction the measurement exceeds by 32% to 141%, the gap growing without bound in w. The residual is signed and interpretable. On the saturated branch the measurement exceeds the prediction by $\kappa = 9$, 86, 109, 133 and 619 steps at increasing Γ: this is the reflection at zero, which invalidates the Wald identity $\mathbb{E}[\tau] = \lambda/\mathbb{E}[\xi]$ under which the deterministic crossing time *is* the expected delay for a free walk. Near the crossover the two agree to 2–4%; on the far ramp branch ($w > 10\,w^*$) the measurement falls 5–11% below, the ordinary first-passage effect of a noisy path crossing ahead of its mean. κ is therefore not the Γ-free constant of Eq. (6), and it is not a power law in Γ either: at $\Gamma = 20$ the delay's standard deviation reaches 3.2× its mean with 1.3% censoring, so a handful of trajectories out of 400 that stay tens of thousands of steps below the barrier—low-volatility stretches in which the standardized squared statistic never clears δ_P—account for most of it. We report κ as an empirical residual and assert no mechanism. Measuring the median instead separates the two effects: MED/ADD falls to 0.68 at the largest penalties, and the median sits 0–18% *below* the deterministic prediction, as the inverse-Gaussian skew of a first-passage law requires. The mean is the right quantity for a detection-delay criterion and the one Eq. (6) approximates; the mean–median gap measures the volatility-clustering tail, not a

failure of the law. Finally, the measured log-log exponent over the ramp branch runs 0.65–0.71, against the 0.71–0.73 that Eq. (6) itself predicts over this range of w: the departure from 1/2 is the dead band, not a failure of the law. Model curves in Figure 5B use the threshold actually applied.

Where the recalibration rule stops holding. A second campaign pushes the horizon to $H = 3 \times 10^6$ and the grid to $225\, w^*$, deep into the dead-band-dominated regime, and locates the boundary of Proposition 2.1 rather than of Eq. (6). The scaling law survives intact where the moment hypothesis is comfortable: at $\Gamma \le 4$ it holds to 5.7% with no fitted constant even at that horizon and width, ρw carrying 78% of the predicted delay and the measured exponents (0.79) matching the model to 1%. The recalibration rule does not. The empirical null quantile departs from $\lambda_{\text{iid}} \times \Gamma$ by +3.3% and +1.1% at $\Gamma = 2$ and 4, but by +27%, +43% and +96% at $\Gamma = 8$, 11.6 and 20. The transition is not gradual in Γ: it coincides with the moment margin δ of the hypothesis $\mathbb{E}|\varepsilon_t|^{4+\delta} < \infty$ falling below 2, i.e. with the loss of $\mathbb{E}[\varepsilon_t^6]$—the second moment of the *monitored* statistic ε_t^2—which for $\alpha = 0.08$ and standardized t_7 innovations occurs at $\Gamma \approx 7.1$. The largest available margin at $\Gamma = 20$ is $\delta \le 0.8$. The underlying effect needs no GARCH at all and is visible at $\Gamma = 1$: over $H = 7.7 \times 10^4$, 2×10^5 and 3×10^6 the i.i.d. threshold grows as $102.8 \to 129.5 \to 303.0$, i.e. as $H^{0.24}$–$H^{0.31}$ rather than the $\log H$ of Siegmund's ARL_0 formula, whose Cramér condition squared innovations do not satisfy. Volatility persistence lowers the tail index further, which is why the error grows with Γ. Two consequences. First, Proposition 2.1 is an asymptotic recipe whose accuracy is governed jointly by the horizon and by the moment margin; we report the main grid at $H = 2 \times 10^5$, inside the regime where it holds to 29%, and state the degradation rather than selecting a horizon that hides it. Second, the fair-coin null of Proposition 3.1 carries *no* moment condition, so no analogue of this boundary exists on the Concept side—an independent reason, beyond the anytime-validity of Section 3, to prefer a monitor whose null law does not have to be re-derived for each horizon. □

D Proof of Proposition 3.1

Proof. Fix t and condition on $\mathcal{F}_{t-1}$. Since $\sigma_t > 0$ is $\mathcal{F}_{t-1}$-measurable and z_t is independent of $\mathcal{F}_{t-1}$ with a continuous distribution symmetric about zero,

$$\mathbb{P}(y_t = 1 \mid \mathcal{F}_{t-1}) = \mathbb{P}(\sigma_t z_t > 0 \mid \mathcal{F}_{t-1}) = \mathbb{P}(z_t > 0) = \tfrac{1}{2}\,.$$

The prediction $\widehat{y}_t$ is $\mathcal{F}_{t-1}$-measurable, hence

$$\mathbb{P}(e_t^{\text{bin}} = 1 \mid \mathcal{F}_{t-1}) = \widehat{y}_t\, \mathbb{P}(y_t = 0 \mid \mathcal{F}_{t-1}) + (1-\widehat{y}_t)\, \mathbb{P}(y_t = 1 \mid \mathcal{F}_{t-1}) = \tfrac{1}{2}\,.$$

The conditional law of e_t^{bin} given $\mathcal{F}_{t-1}$ is thus the fixed Bernoulli(1/2) law, independent of the past. Since $(e_1^{\text{bin}}, \dots, e_{t-1}^{\text{bin}})$ is $\mathcal{F}_{t-1}$-measurable, e_t^{bin} is independent of $(e_1^{\text{bin}}, \dots, e_{t-1}^{\text{bin}})$ with a constant marginal. By induction, (e_t^{bin}) is i.i.d. Bernoulli(1/2). □

E Sharpe–KL Conversion for Corollary 3.6

For daily Gaussian returns, $\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, \sigma^2)\big) = \mu^2/(2\sigma^2) = \mathrm{SR}_{\text{daily}}^2/2 = \mathrm{SR}^2/504$, since $\mathrm{SR} = \sqrt{252}\,\mathrm{SR}_{\text{daily}}$. The lower bound $\mathrm{ADD}_{\min} \ge \ln(\gamma)/\mathrm{KL}\,(1 + o(1))$ is the first-order information bound of quickest detection [36]; the Pitman efficiency used in Proposition 3.5 is classical [38].

F Supplementary Figures

This appendix collects the figures supporting the robustness and detector-generalization results whose quantitative conclusions are stated in the main text.

Table 3: Iso-FPR race under a location drift ($\Gamma \approx 11.6$, standardized Student-t_{30} innovations, all arms calibrated to 5% FPR, 2,000 streams). ADD in steps; DetRate in parentheses when below 1.

c	Recalib ($(\varepsilon/\hat{\sigma})^2$)	Eco-L1 ($\varepsilon/\hat{\sigma}$)	Concept (sign)
0.25	2293 (0.19)	389	460
0.50	1337 (0.89)	72.0	101
1.00	203	26.4	43.8
2.00	55.9	12.6	28.9

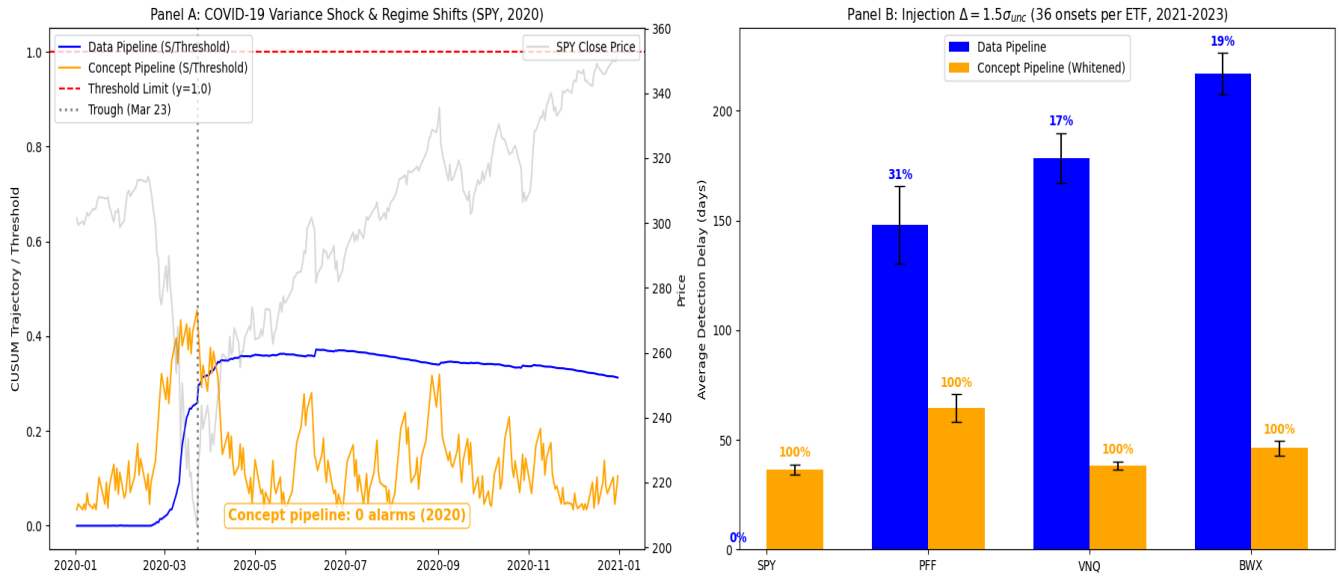


Figure 2: Real-world validation on ETF streams (FirstRate Data, SPY). (A) Normalized CUSUM S_t/λ through the 2020 crash: DATA is throttled by its Γ-inflated threshold (Lethargy Tax); CONCEPT exhausts its budget on a crash too brief to afford. (B) Directional injection ($1.5\sigma_{\text{unc}}$, 2021–2023) with placebo control: squared sensor is weak and placebo-confounded, while sign filter achieves 100% detection (ADD 36.6–64.6 days).

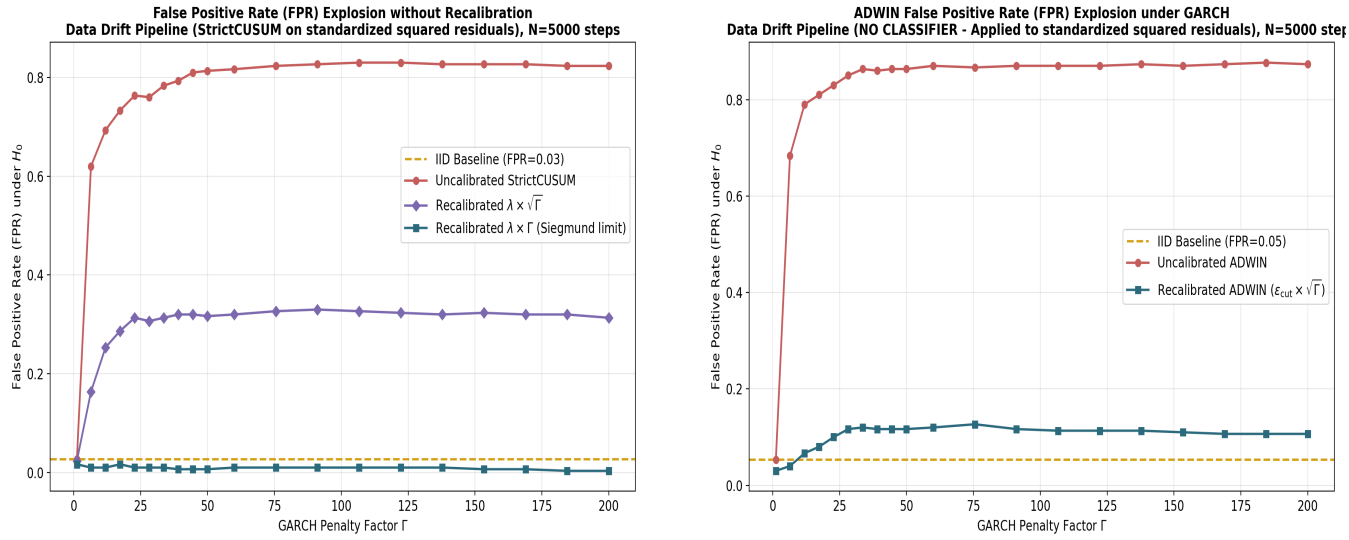


Figure 3: The FPR explosion and its detector-dependent cure (H_0 GARCH streams, DATA statistic). (A) StrictCUSUM: uncalibrated FPR explodes above 80%; $\lambda \times \sqrt{\Gamma}$ plateaus near 30%; the full $\lambda \times \Gamma$ inflation holds the nominal 5%. (B) ADWIN: the window-mean cut scales with the standard deviation, so $\epsilon_{\text{cut}} \times \sqrt{\Gamma}$ contains the FPR within $\approx 2.5\times$ nominal.

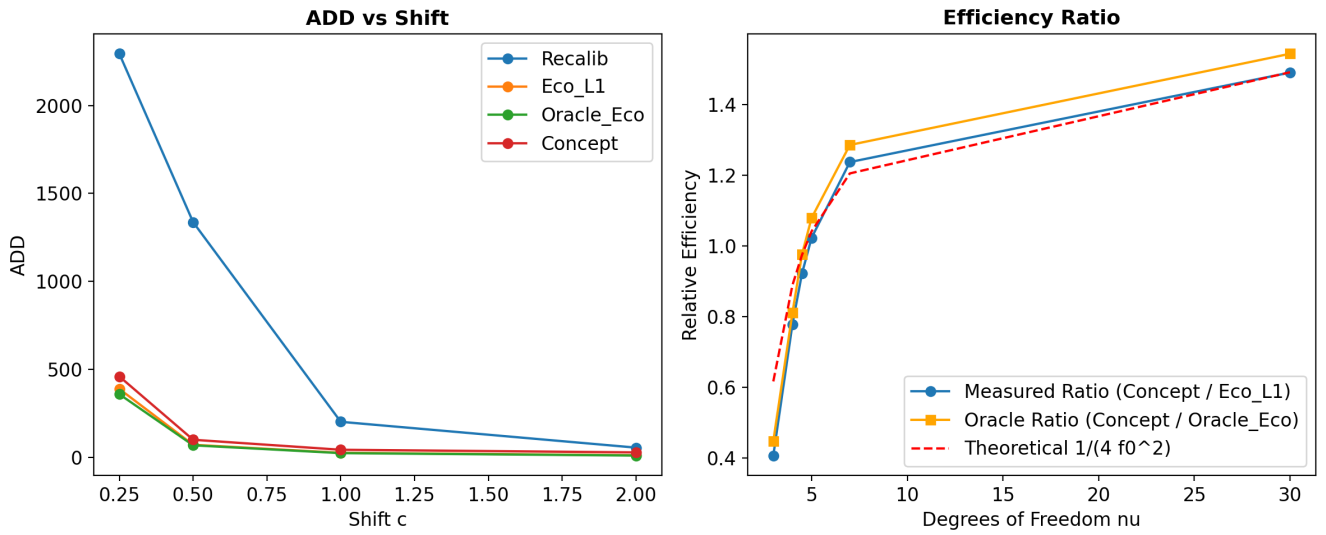


Figure 4: Iso-FPR race and relative efficiency (all arms calibrated to 5% FPR by bisection, 2,000 null streams). (A) Delay vs. drift magnitude c at $\Gamma \approx 11.6$: the squared sensor is 2–19× slower and collapses below $c \approx 0.5$. (B) $\mathrm{ADD}_{\mathrm{Concept}}/\mathrm{ADD}_{\mathrm{Eco\text{-}L1}}$ vs. ν: the ratio matches the predicted crossover of $1/(4f_z(0)^2)$, crossing unity at a measured $\nu^\star \approx 4.9$ (analytic 4.7 for the exactly standardized ideal), below which the sign filter overtakes parametric standardization.

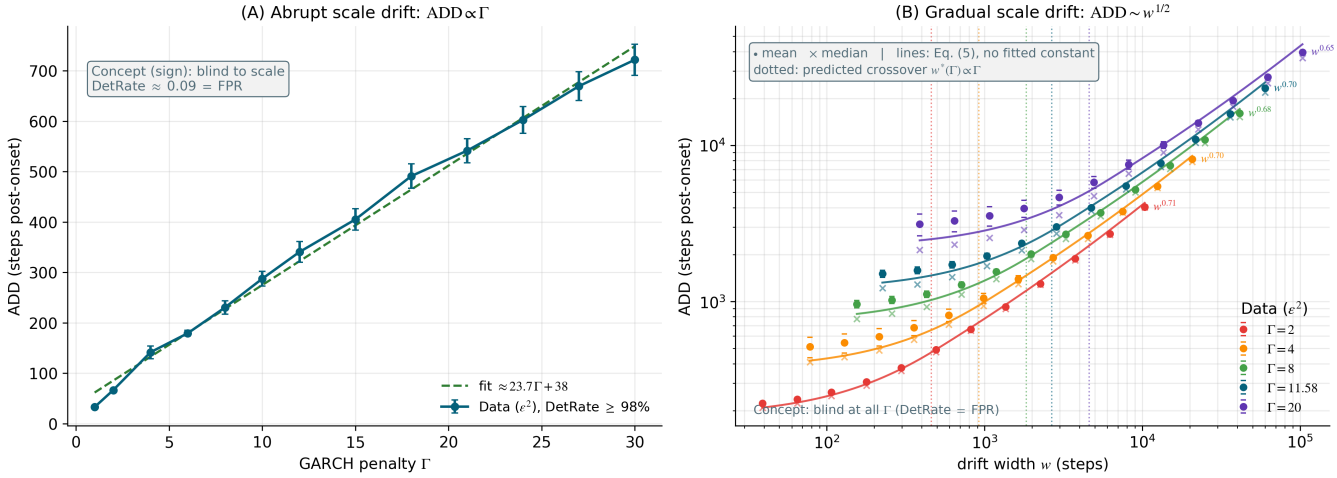


Figure 5: Scale-monitoring law and location/scale orthogonality (5% false alarms per Γ, 400 seeds; error bars = SEM). (A) Abrupt scale shift: DATA delay is linear in Γ ($\mathrm{ADD} \approx 23.7\,\Gamma + 38$). (B) Gradual ramps across five penalties $\Gamma \in \{2, \ldots, 20\}$ up to $w = 22.5\,w^*(\Gamma)$. Solid lines show Theorem 2.2 (Eq. (6)) with no fitted constant; dotted verticals mark the crossover $w^*(\Gamma)$. Filled markers: mean delay; crosses: median. In both panels, the sign monitor remains blind to scale drifts (detection rate = false-alarm rate, Prop. 3.3).

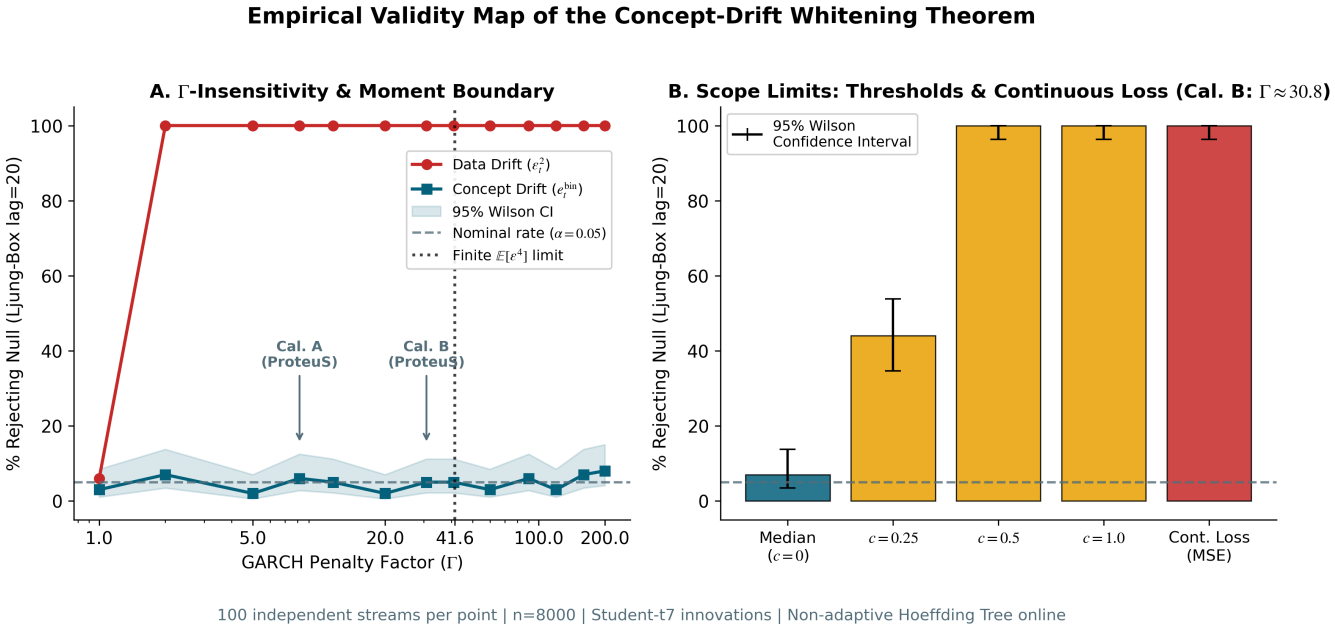


Figure 6: Empirical validity map of the whitening property (100 streams per configuration). (A) Γ-insensitivity holds past the loss of the fourth moment ($\Gamma \approx 41.6$). (B) Shifted threshold ($c > 0$) or continuous MSE loss breaks whitening, mapping the boundaries of Remark 3.2.

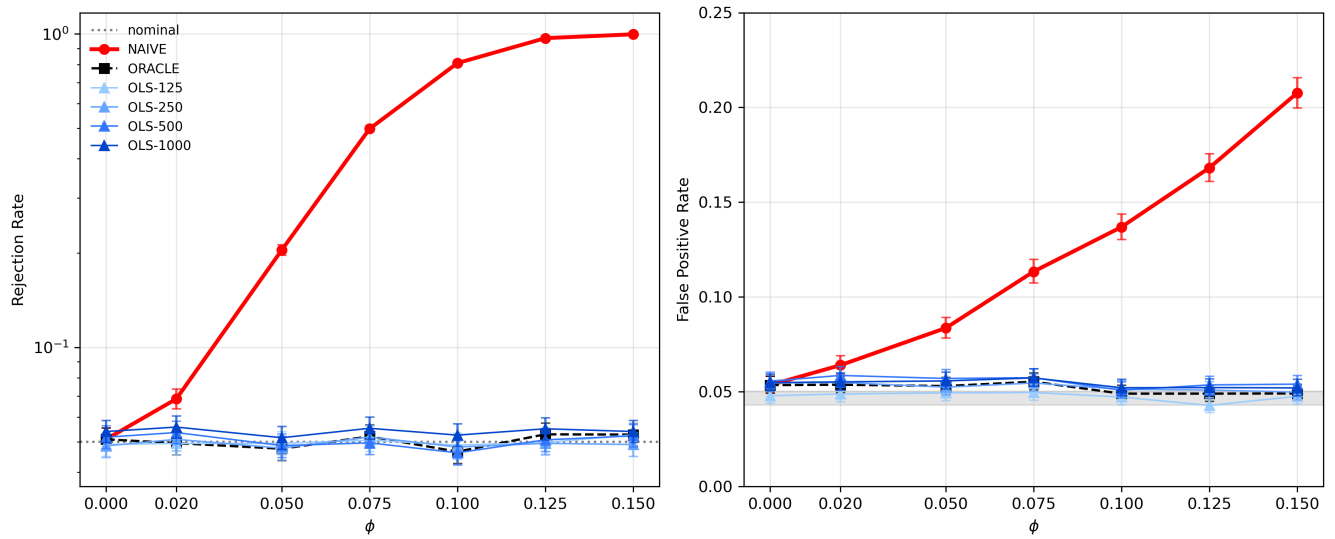


Figure 7: Whitening under an estimated conditional mean (10,000 paired AR(1)-GARCH trajectories per ϕ, horizon $H = 5{,}000$; error bars = Wilson intervals). (A) Ljung–Box rejection of the sign stream vs. momentum ϕ. (B) Concept FPR at threshold $\lambda^\star = 11.4$ (exact lattice level 4.29%, 5.03% at $\lambda = 11.2$; shaded band = the two attainable levels bracketing 5%). Naive recentering ($\hat{\mu}_t = 0$) degrades rapidly (FPR reaching 20.8% at $\phi = 0.15$), while all rolling-OLS arms ($n \in [125, 1000]$) match oracle false-alarm control. Calibration depends on estimator *bias*, not dispersion.

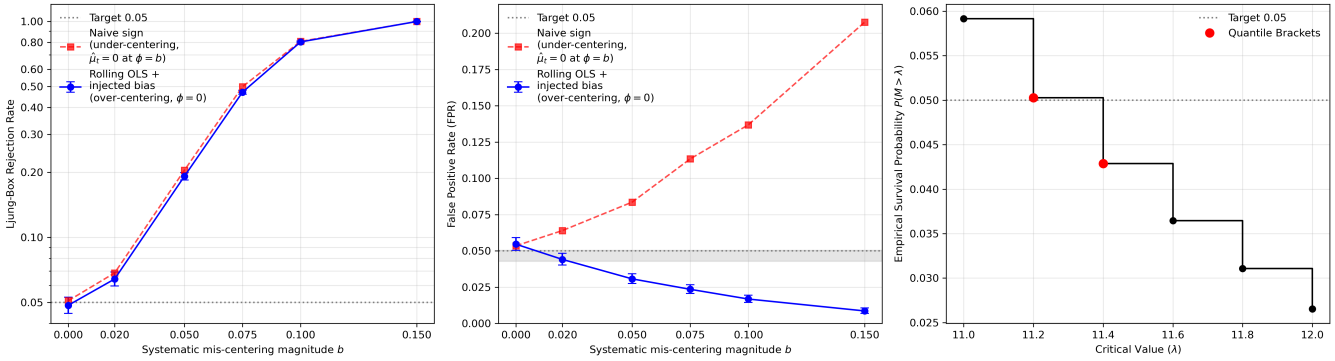


Figure 8: The adverse direction and discrete null law (10,000 trajectories/point for A–B; 2×10^5 fair-coin streams for C). (A) Ljung–Box rejection: injected over-centering bias b and naive under-centering at $\phi = b$ yield identical whiteness loss ($|\mathrm{Cov}(y_t, y_{t+k})|$ response). (B) False-alarm rate: over-centering induces negative autocorrelation (FPR collapses to 0.86%), whereas under-centering induces positive autocorrelation (FPR inflates to 20.8%). (C) Survival function of M_H under H_0: step function on multiples of 2δ with the two levels bracketing 5% marked.

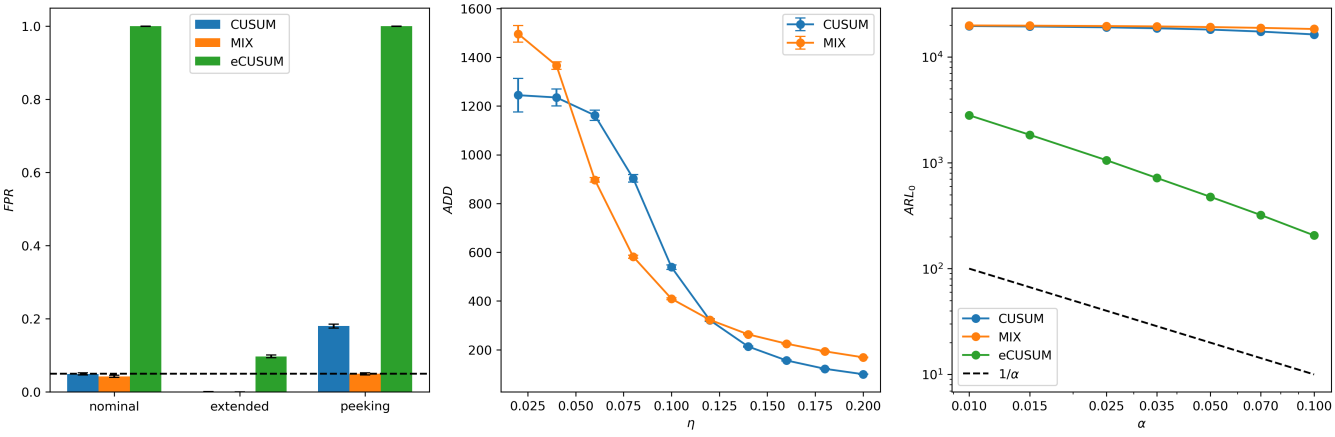


Figure 9: Anytime-valid detection on the fair-coin stream (2×10^4 streams per cell). (A) Realized false-alarm rate under continuous monitoring (peeking over $[1, 4H]$): fixed-horizon CUSUM climbs to 18%, whereas the mixture martingale (MIX) remains bounded by α. (B) Detection delay vs. drift η ($\alpha = 0.05$): MIX matches CUSUM speed for moderate drifts ($\eta \leq 0.10$). (C) Average run length vs. α: e-CUSUM satisfies $\mathrm{ARL}_0 \geq 1/\alpha$. Only MIX controls the time-uniform false-alarm probability.

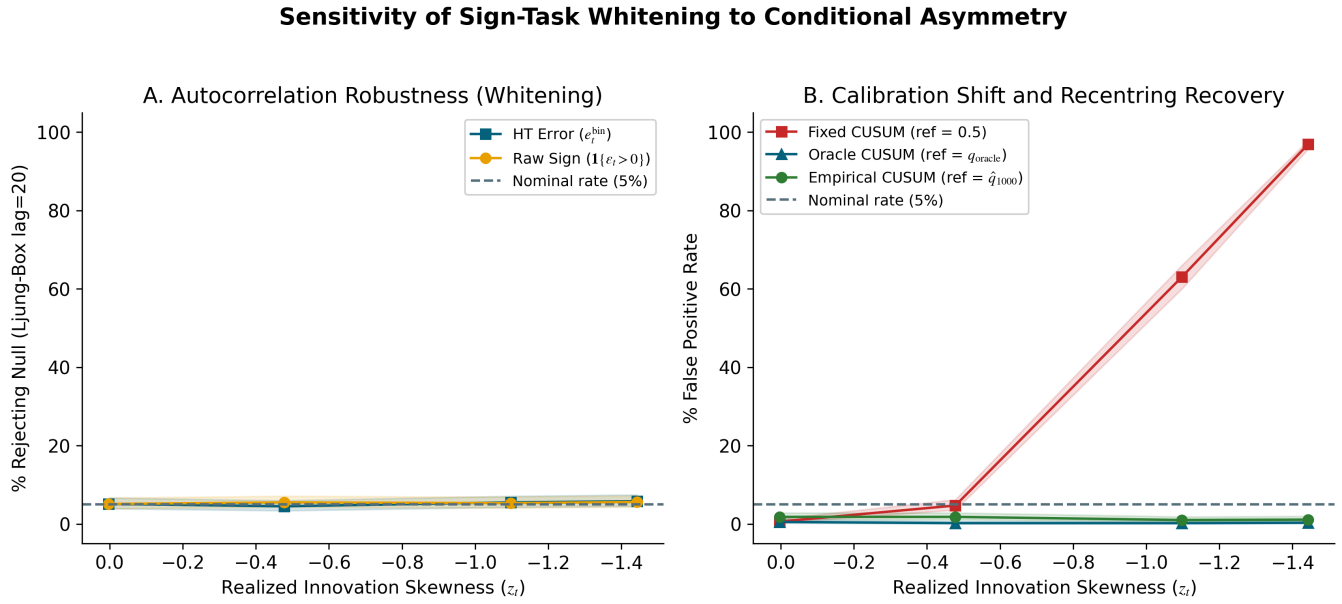


Figure 10: Sensitivity to conditional asymmetry (1,000 streams per point). (A) Conditional whiteness is preserved across extreme skewness. (B) Asymmetry shifts the marginal rate; recentering on warm-up estimate $\hat{q}$ restores false-alarm control (measured FPR 1.0–1.8%).

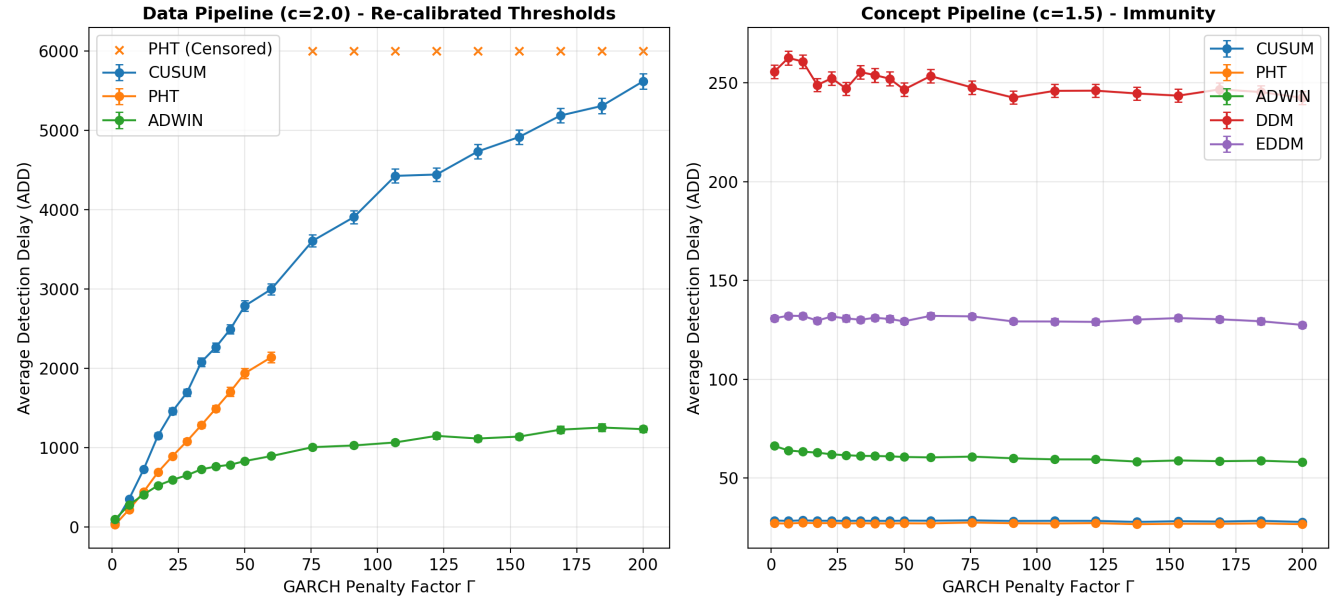


Figure 11: The Lethargy Tax vs. GARCH immunity (abrupt drift $c = 2$, 1,000 streams per point). (A) Raw Data: the recalibrated delay grows with Γ at each detector's own recalibration exponent; PHT collapses near $\Gamma \approx 75$ (stochastic syncope). (B) Whitened Concept: flat delays for all detectors.

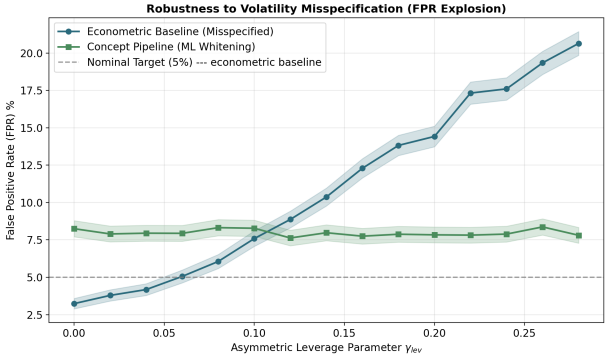


Figure 12: Robustness to volatility misspecification (10,000 **streams/point, pseudo-Gaussian innovations). Symmetric GARCH standardization on GJR-GARCH leverage inflates the baseline's Ljung–Box rejection** (5.1% → 24.6%) **and drives its FPR from** 3.2% **to** 20.6%**; the sign pipeline holds a leverage-invariant false-alarm rate** (7.6–8.4% **at its fixed threshold), its null law is Bernoulli**$(1/2)$ **(Corollary 3.4).**

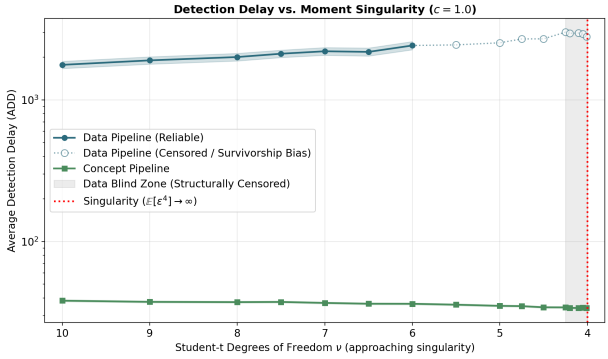


Figure 13: Detection delay vs. moment singularity (1,000 **streams/point). As** $\nu \to 4$**, the** Data **pipeline collapses (detection** $< 50\%$**) due to divergent sample variance, whereas** Concept **delay remains flat.**

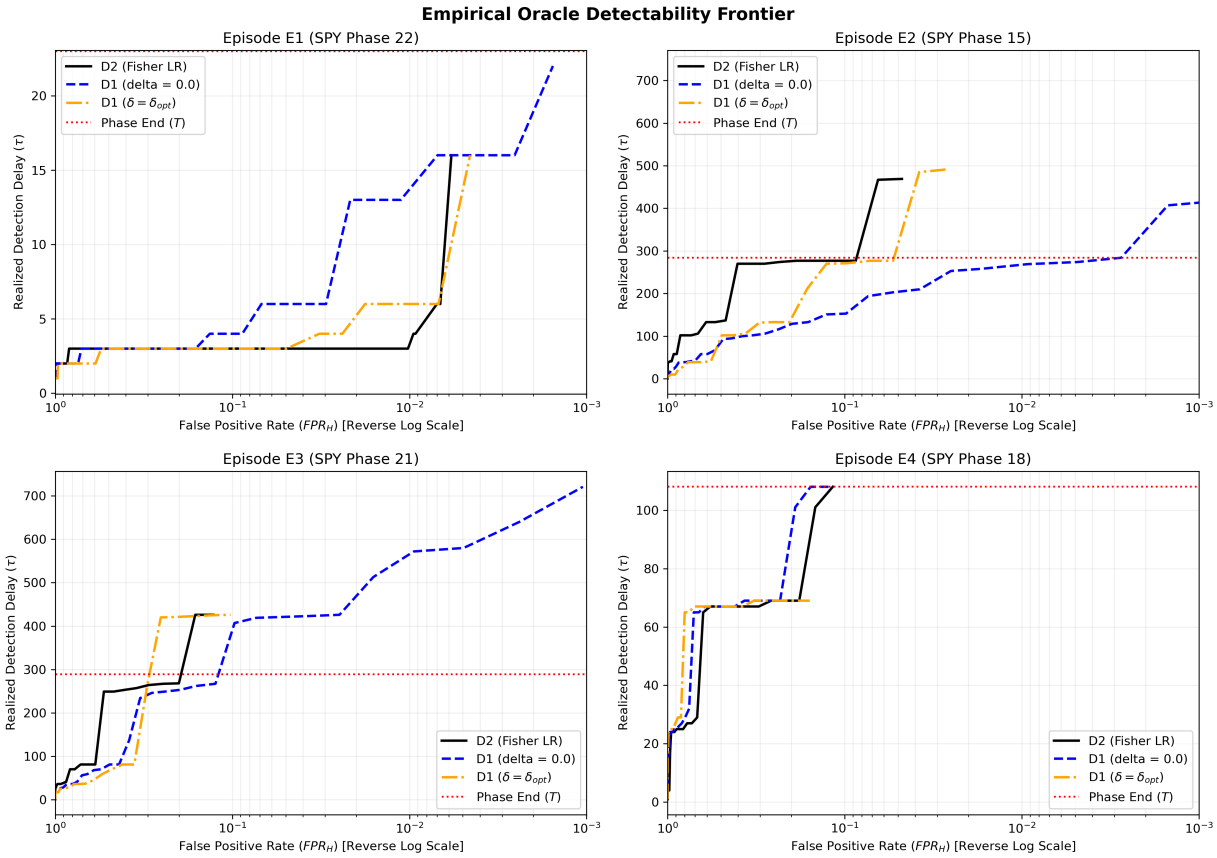


Figure 14: Empirical oracle detectability frontier (look-ahead GARCH parameters, causal filtration): realized delay τ **vs. bootstrap** FPR_H **under a frozen-volatility null, for the likelihood-ratio CUSUM and the standardized-mean CUSUM (**$\delta = 0$ **and** δ_{opt}**). Dotted line: phase end (**T**). The 2020 crash is detected well before** T **at sub-percent false alarms; the 2011 correction is not detected at either setting.**

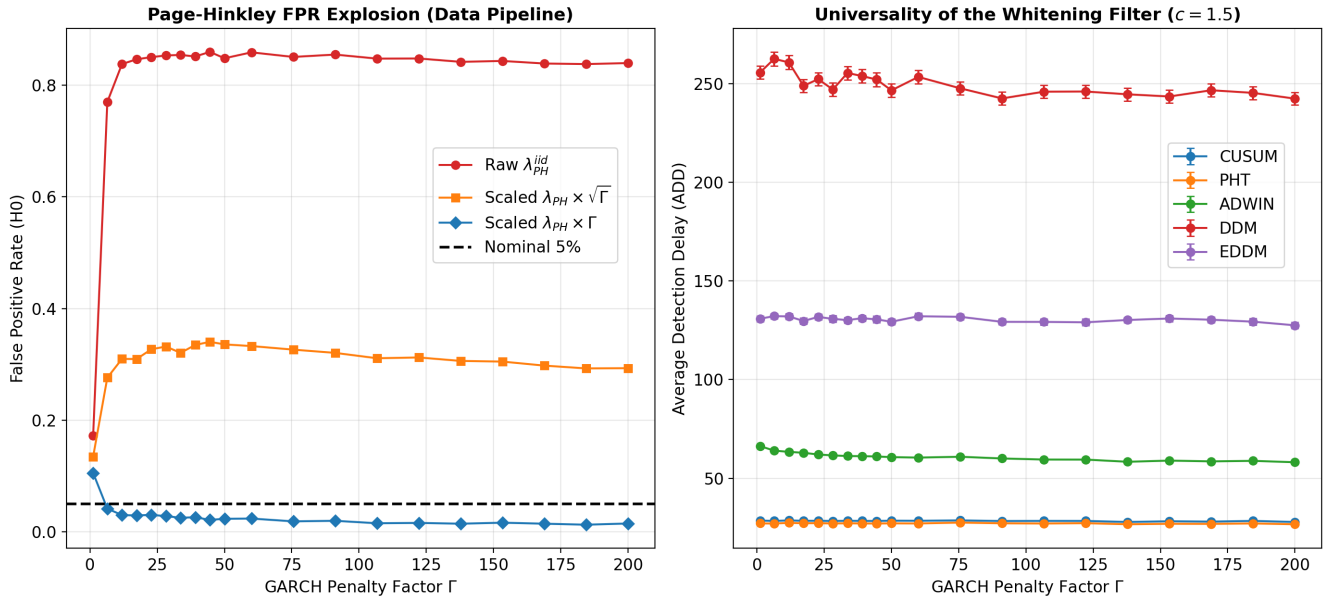


Figure 15: Multi-detector generalization. (A) The Page–Hinkley Test is cumulative like the CUSUM: same FPR explosion, same $\lambda \times \Gamma$ cure, same ≈30% plateau under $\sqrt{\Gamma}$. (B) Universality of the whitening filter ($c = 1.5$): flat delays across CUSUM (≈ 28.3), PHT (≈ 27.1), ADWIN (≈ 61), DDM (≈ 250); EDDM's flat curve is a noise artifact of its incompatibility with maximum-entropy streams.

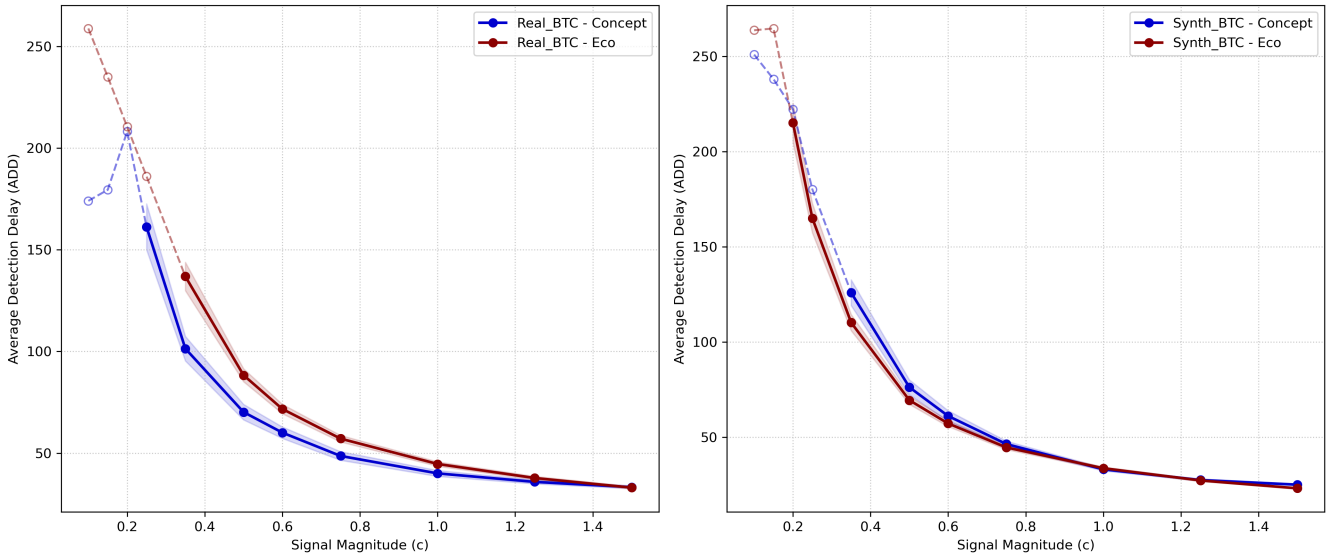


Figure 16: Efficiency reversal on real Bitcoin (iso-FPR 4.7% on real placebo windows, 106 monthly onsets; hollow markers are unreliable points, DetRate < 0.9). (A) Daily BTC ($\hat{\nu} = 2.78$): the sign filter (CONCEPT) leads the honestly standardized residual (ECO-L1) across the reliable range, converging to parity at $c = 1.5$. (B) A quasi-Gaussian t_{30} control on the same onsets inverts the ordering, reproducing the light-tailed Pitman penalty of Figure 4B on real machinery.

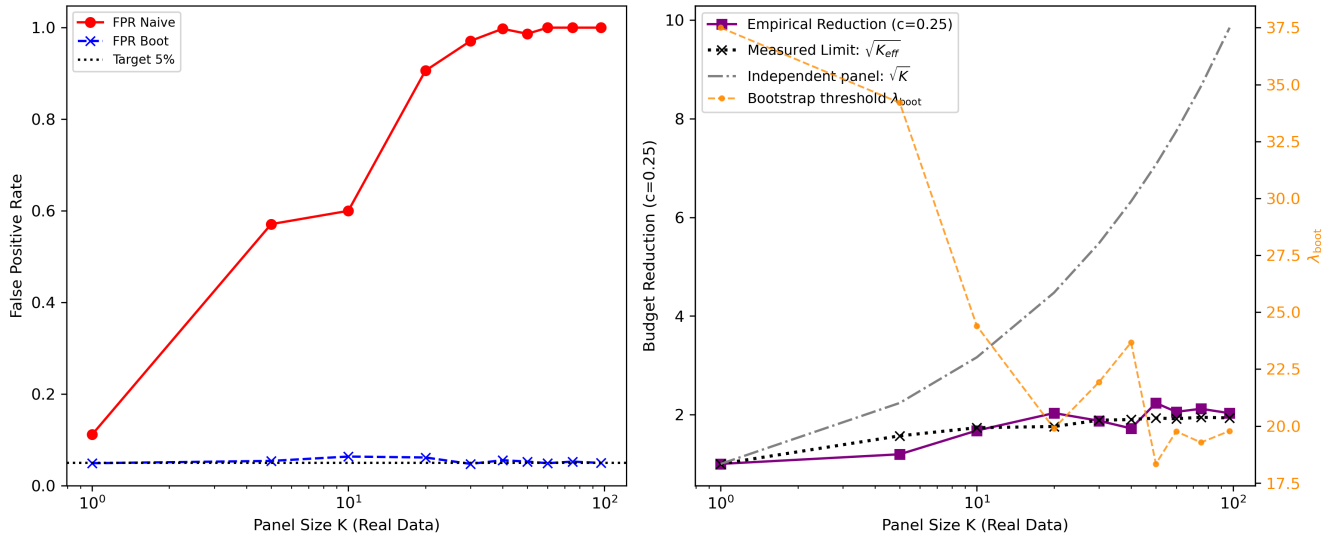


Figure 17: Cross-sectional escape on a real equity panel (97 surviving US equities, 2005–2025, upper-bound co-movement benchmark). (A) An independence calibration lets false alarms climb toward 100% by ignoring sign correlation ($\hat{\rho} \approx 0.26$), whereas a real-window bootstrap holds the nominal 5% level (4.8–6.4%). (B) Realized budget reduction plateaus near 2× ($K \geq 20$), bounded by the effective panel size $K_{\text{eff}} \approx 1/\hat{\rho} \approx 3.8$. Point-to-point scatter reflects threshold variations across panel compositions ($r \geq 0.99$ with bootstrap threshold).